

TSUOV:Tensor Unbalanced Oil and Vinegar

Algorithm Specifications and Supporting Documentation

Version 1.0

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Contents

1	Introduction	3
1.1	Background and motivation	3
1.2	TSUOV	4
2	Preliminaries	4
2.1	Notation	4
2.2	Kronecker product	5
2.3	Hard problems in multivariate cryptography	6
2.3.1	The cost of solving MQ problem	6
2.3.2	The cost of solving Minrank problem	7
2.4	Description of UOV and its variants	7
2.4.1	UOV	7
2.4.2	MAYO	9
2.4.3	QR-UOV	10
2.4.4	SNOVA	10
3	Design rationale	11
3.1	Tensor structure of UOV variants	11
3.1.1	MAYO	11
3.1.2	QR-UOV	13
3.1.3	SNOVA	13
4	Algorithm Specification	14
4.1	Algorithm Description	14
4.2	Representation of Keys and Signature	17
4.3	Basic functionalities	18
4.4	Parameter sets	21
5	Performance Analysis	22
5.1	Reference implementation	22
5.2	Performance-optimized implementation	23
6	Security Analysis	24
6.1	Hard Problems underlying TSUOV	24
6.2	Security Proof	24
6.3	Analysis of known attacks	30
6.3.1	Forgery attack	30
6.3.2	Attacks on the Oil and Vinegar problem	32
7	Advantages and Limitations	37
7.1	Advantages	37
7.2	Limitations	38

1 Introduction

This document gives a detailed description of TSUOV (Tensor UOV), a multivariate quadratic signature scheme that uses tensor product language to generalize three UOV variants that are third round candidates in NIST additional signature standardization process.

1.1 Background and motivation

Public key cryptosystems currently in widespread use, such as RSA and elliptic curve cryptography (ECC), are vulnerable to quantum attacks: Shor’s algorithm can factor large integers and compute discrete logarithms in polynomial time [34]. This motivates the urgent search for post-quantum cryptography (PQC). Candidate systems arise from several mathematical foundations, including lattices, error-correcting codes, algebraic geometry, and isogeny theory.

To guide this process, the U.S. National Institute of Standards and Technology (NIST) has been coordinating an international standardization effort. After the third round of the PQC competition, NIST issued in 2022 a call for further digital signature proposals. The following year, more than 50 candidates were submitted, covering lattice-based, code-based, isogeny-based, and multivariate schemes, among others. In October 2025, the Institute of Commercial Cryptography Standards of China also launched a call for proposals on post-quantum cryptography schemes.

Among these, multivariate public key cryptosystems (MPKCs) are especially attractive because of their efficiency and compact signatures. An MPKC publishes a collection of quadratic polynomials over a finite field, and its security depends on the difficulty of solving a system of such equations, known as the multivariate quadratic (MQ) problem. The MQ problem has been proven NP-complete in general [21], and has underpinned a long line of cryptographic constructions.

Several notable MPKC schemes have demonstrated impressive resilience. The unbalanced oil and vinegar (UOV) family [30, 26], has remained secure for over two decades. The Rainbow scheme [12], a multilayer UOV construction, advanced to the third round of the NIST PQC project. Although specific Rainbow parameter sets were broken [5], the structural core of UOV is still considered robust. In contrast, some other multivariate designs, such as HFE [29], eventually succumbed to algebraic cryptanalysis [37]. A well-known drawback of UOV-style systems, however, is their large public key sizes compared to lattice-based or hash-based signatures.

To reduce the public-key size of UOV variants, researchers have proposed a range of strategies that intervene at different stages of the scheme’s design.

1. Key generation compression. One line of work does not alter the UOV structure itself, but reduces redundancy in the key material. For example, Petzoldt et al. [32] showed that parts of the public key can be deterministically regenerated from a small seed, shrinking storage requirements while leaving the signing process unchanged.

2. Working over small fields with extension mappings. Another approach is to define the public polynomials over a small base field, while the message and signature spaces live in an extension field. This principle underlies LUOV [8], although some of its proposed parameters were later broken [13].
3. Expanding the dimension of oil space. A more recent family of techniques reduces the explicit oil dimension at key generation and then expands it during signing by embedding or lifting constructions:
 - QR-UOV [19] creates oil spaces over extension fields and maps them back to the base field using the trace or tensor product, producing sufficiently large effective oil spaces. Algebraically, this again amounts to embedding vectors into a larger structured oil space via Kronecker-type products. Related schemes such as BAC-UOV [36] follow a similar spirit but have been cryptanalyzed [20].
 - MAYO [7] enlarges the vinegar-oil map $\mathcal{P}$ into a higher-dimensional map $\mathcal{P}^*$, again effectively boosting the oil space.
 - SNOVA [39] introduces a noncommutative coefficient ring of matrices and constructs signatures in a matrix ring setting.

1.2 TSUOV

In [28], the authors propose a unified approach to analyze and enhance a class of multivariate signature schemes derived from the UOV scheme. Although these approaches are often presented as distinct design philosophies, [28] emphasize that they share a common underlying principle: each effectively expands the dimension of the oil space through tensor-like constructions. Whether realized via trace maps, block embeddings, matrix ring structures, or whipping techniques, these schemes rely on higher-dimensional lifts to balance the number of equations against the available oil variables.

Building upon this understanding, we introduce a more general multivariate signature scheme, named TSUOV, which integrates the advantages of QR-UOV, MAYO, and SNOVA, while providing security guarantees no weaker than those of these individual schemes. This demonstrates the broader potential of our framework, both in clarifying the design space of UOV-like schemes and in guiding the development of improved post-quantum signature constructions. As a result, TSUOV achieves a significant reduction in public key size compared to existing UOV variants, while maintaining competitive signature sizes and computational efficiency. Furthermore, TSUOV’s unified design allows for flexible parameter selection and potential future enhancements, making it a promising candidate for practical deployment in the post-quantum era.

2 Preliminaries

2.1 Notation

In this document, we use the following notation:

- $n, o, m_1, m_2, q, \ell, k$ are positive integers defining the parameters of TSUOV.
- λ , security parameter.
- $[c]$, the set $\{1, \dots, c\}$.
- $\mathbb{F}_q$ denotes a finite field with q elements.
- $\text{char}(\mathbb{F}_q)$ denotes the characteristic of the field $\mathbb{F}_q$.
- $\mathbb{F}_q^c$ denotes the space of c -dimensional column vectors over $\mathbb{F}_q$.
- $\text{Mat}_{r \times c}(\mathbb{F}_q)$ denotes the set of matrices of size $r \times c$ with entries in $\mathbb{F}_q$.
- $A^\top$ the transpose of the matrix A .
- $\otimes$ denotes the Kronecker product.
- $\text{UOV}(n, o, m, q)$ a UOV scheme with n variables, o oil variables, and m quadratic equations over $\mathbb{F}_q$.
- $\Lambda_{\mathbf{Q}}^{(n)}$ is the block diagonal matrix with n copies of $\Lambda_{\mathbf{Q}}$ on the diagonal.

2.2 Kronecker product

If $\mathbf{A}$ is an $m \times n$ matrix and $\mathbf{B}$ is a $p \times q$ matrix, then the Kronecker product $\mathbf{A} \otimes \mathbf{B}$ is the $pm \times qn$ block matrix:

$$\begin{bmatrix} a_{1,1}\mathbf{B} & \cdots & a_{1,n}\mathbf{B} \\ \vdots & \ddots & \vdots \\ a_{m,1}\mathbf{B} & \cdots & a_{m,n}\mathbf{B} \end{bmatrix}$$

From fundamental matrix textbooks such as [24], one can find proofs for the following proposition and lemma.

Proposition 2.1. *Let $\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}$, $\mathbf{D}$ be given matrices, we have*

1. *The Kronecker product is a special case of the tensor product, so it is bilinear and associative:*

$$\begin{aligned} \mathbf{A} \otimes (\mathbf{B} + \mathbf{C}) &= \mathbf{A} \otimes \mathbf{B} + \mathbf{A} \otimes \mathbf{C} \\ (\mathbf{B} + \mathbf{C}) \otimes \mathbf{A} &= \mathbf{B} \otimes \mathbf{A} + \mathbf{C} \otimes \mathbf{A} \\ (k\mathbf{A}) \otimes \mathbf{B} &= \mathbf{A} \otimes (k\mathbf{B}) = k(\mathbf{A} \otimes \mathbf{B}) \\ (\mathbf{A} \otimes \mathbf{B}) \otimes \mathbf{C} &= \mathbf{A} \otimes (\mathbf{B} \otimes \mathbf{C}) \\ \mathbf{A} \otimes \mathbf{0} &= \mathbf{0} \otimes \mathbf{A} = \mathbf{0} \end{aligned}$$

where k is a scalar, $\mathbf{0}$ is a zero matrix,

2. $(\mathbf{A} \otimes \mathbf{B})^\top = \mathbf{A}^\top \otimes \mathbf{B}^\top$.

3. If $\mathbf{C}$ and $\mathbf{D}$ are matrices of proper size such that one can form the matrix products $\mathbf{AC}$ and $\mathbf{BD}$, then $(\mathbf{A} \otimes \mathbf{B}) \cdot (\mathbf{C} \otimes \mathbf{D}) = (\mathbf{AC}) \otimes (\mathbf{BD})$.
4. $\mathbf{A} \otimes \mathbf{B}$ is invertible if and only if both $\mathbf{A}$ and $\mathbf{B}$ are invertible, in which case the inverse is given by $(\mathbf{A} \otimes \mathbf{B})^{-1} = \mathbf{A}^{-1} \otimes \mathbf{B}^{-1}$.

Lemma 2.2. Consider for instance the equation $\mathbf{AXB} = \mathbf{C}$ where $\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}$ and $\mathbf{X}$ are given matrices, we have

$$(\mathbf{B}^\top \otimes \mathbf{A}) \cdot \text{vec}(\mathbf{X}) = \text{vec}(\mathbf{C}) \quad (1)$$

where $\text{vec}(\mathbf{X})$ is the vectorization of the matrix $\mathbf{X}$, that is, the column vector obtained by stacking the columns of $\mathbf{X}$ on top of one another.

2.3 Hard problems in multivariate cryptography

2.3.1 The cost of solving MQ problem

Definition 2.3 (MQ problem). Given a multivariate quadratic map (not necessarily homogeneous) $\mathcal{P}: \mathbb{F}_q^N \rightarrow \mathbb{F}_q^M$, the MQ problem is to find a solution $\mathbf{u}$ satisfying $\mathcal{P}(\mathbf{u}) = 0$ if such solution exists.

We use $\text{MQ}(N, M, q)$ to denote the complexity of finding a solution $\mathbf{u}$ satisfying $\mathcal{P}(\mathbf{u}) = 0$ if such solution exists. Several algorithms for algebraically solving the quadratic system by computing Gröbner basis [10] include F_4 [15], F_5 [16] and XL [11]. One of the best-known approaches for solving the quadratic system is the hybrid approach [3], which randomly guesses k ($k = 0, \dots, N$) variables before computing a Gröbner basis. The complexity of this approach is given as

$$\text{MQ}(N, M, q) = \min_k q^k \cdot 3 \binom{N - k + d_{\text{reg},k}}{d_{\text{reg},k}}^2 \binom{N - k + 2}{2} \quad (2)$$

field multiplications, where $d_{\text{reg},k}$ is equal to the degree of the first non-positive term in the series generated by

$$\frac{(1 - t^2)^M}{(1 - t)^{N-k+1}}.$$

By using Grover's algorithm [22], the complexity of a quantum direct attack is estimated as

$$\text{MQ}_{\text{quantum}}(N, M, q) = \min_k q^{k/2} \cdot 3 \binom{N - k + d_{\text{reg},k}}{d_{\text{reg},k}}^2 \binom{N - k + 2}{2}. \quad (3)$$

When $N > M$, Thomae and Wolf [38] proposed a technique to reduce such quadratic system to another quadratic system with $M - \alpha + 1$ variables and equations, where $\alpha = \lfloor N/M \rfloor$. Furthermore, when

$$N \geq \max\{(\alpha + 1)(M - k - \alpha + 1), \alpha(M - k) - (\alpha - 1)^2 + k\}$$

Hashimoto [23] found a solution to $\mathcal{P}(\mathbf{s}) = \mathbf{t}$ with an estimated cost of

$$\min_{\alpha, k} \{ (M - \alpha - k + 1) \text{MQ}(\alpha, \alpha, q) + q^k (\text{MQ}(\alpha - 1, \alpha - 1, q) + \text{MQ}(M - \alpha - k, M - \alpha, q)) \}$$

2.3.2 The cost of solving Minrank problem

Definition 2.4 (Minrank problem). Let $\mathbf{M}_1, \dots, \mathbf{M}_k$ be k matrices of size $n \times m$ with $n \geq m$. Let r be an integer satisfying $r < m$. The Minrank problem is the problem of finding nontrivial coefficients $c_1, \dots, c_k$ such that

$$\text{rank}(c_1 \mathbf{M}_1 + \dots + c_k \mathbf{M}_k) \leq r.$$

One of the algorithms for solving the Minrank problem is the support minors method [1]. The idea is to reduce the Minrank problem to a bilinear quadratic system. The complexity is given by

$$3M(b_{\min}, 1)^2(r+1)k,$$

where $M(b, 1) = \binom{m}{r} \binom{k+b-1}{b}$ is the number of columns in the Macaulay matrix $\mathbf{M}(b, 1)$ of bidegree $(b, 1)$ and $b_{\min}$ is the smallest value b satisfying

$$M(b, 1) - 1 < R(b) := \sum_{i=1}^b (-1)^{i+1} \binom{m}{r+i} \binom{n+i-1}{i} \binom{k+b-i-1}{b-i}.$$

2.4 Description of UOV and its variants

2.4.1 UOV

In the (unbalanced) Oil and Vinegar signature scheme [30], we separate the variables $\mathbf{x} = (x_1, \dots, x_n)^\top$ over $\mathbb{F}_q$ as two parts: the variables $x_1, \dots, x_v$ are called vinegar variables, and the variables $x_{v+1}, \dots, x_n$ are called oil variables. The central map is represented by a trapdoor homogeneous multivariate map $\mathcal{F} = (f_1, \dots, f_m): \mathbb{F}_q^n \rightarrow \mathbb{F}_q^m$ such that each f_k for $k \in [m]$ is a quadratic polynomial of the form

$$f_k(x_1, \dots, x_n) = \sum_{i=1}^v \sum_{j=i}^n \alpha_{i,j}^{(k)} x_i x_j \quad (4)$$

for some $\alpha_{i,j}^{(k)} \in \mathbb{F}_q$. Since there are no crossing terms of oil variables, it is obvious from definition that $\mathcal{F}$ has an isotropic subspace $\mathcal{O}_0 = \{0\}^v \times \mathbb{F}_q^{n-v}$ (which means $\mathcal{F}|_{\mathcal{O}_0} = 0$).

We will also choose a random linear map $\mathcal{S}: \mathbb{F}_q^n \rightarrow \mathbb{F}_q^n$ to hide the structure of $\mathcal{F}$. The public key map $\mathcal{P}$ is the composition $\mathcal{F} \circ \mathcal{S}: \mathbb{F}_q^n \rightarrow \mathbb{F}_q^m$ (namely $p_i(\mathbf{x}) = f_i(\mathcal{S}(\mathbf{x}))$), and the private key is the pair $(\mathcal{F}, \mathcal{S})$. $\mathcal{O} := \mathcal{S}^{-1}(\mathcal{O}_0)$ is called the oil subspace of $\mathcal{P}$. Note that $\dim_{\mathbb{F}_q} \mathcal{O} = o := n - v$ and for any $\mathbf{u} \in \mathcal{O}$, we have

$$\mathcal{P}(\mathbf{u}) = 0. \quad (5)$$

If we rewrite the quadratic form in Eq. (4) as a vector acting on both sides of a matrix, then we can write it as

$$f_k(x_1, \dots, x_n) = \mathbf{x}^\top \mathbf{F}_k \mathbf{x} = \begin{bmatrix} x_1 & \dots & x_n \end{bmatrix} \begin{bmatrix} * & & * \\ & \mathbf{0}_{(n-v) \times (n-v)} & \\ * & & \end{bmatrix} \begin{bmatrix} x_1 & \dots & x_n \end{bmatrix}^\top \quad (6)$$

where the lower-right $(n - v) \times (n - v)$ submatrix of $\mathbf{F}_k$ is zero. Similarly we have

$$p_k(x_1, \dots, x_n) = [x_1 \ \dots \ x_n] \mathbf{P}_k [x_1 \ \dots \ x_n]^\top \quad (7)$$

for some square matrix $\mathbf{P}_k$, and

$$\mathcal{S}(x_1, \dots, x_n) = \mathbf{S} [x_1 \ \dots \ x_n]^\top \quad (8)$$

for some square matrix $\mathbf{S}$. As a result we can choose $\mathbf{P}_k$ such that

$$\mathbf{P}_k = \mathbf{S}^\top \mathbf{F}_k \mathbf{S} \quad (9)$$

To further reduce the key size, one can specify the form of $\mathbf{S}$ to be

$$\mathbf{S} = \begin{bmatrix} \mathbf{I}_{v \times v} & * \\ \mathbf{0}_{(n-v) \times v} & \mathbf{I}_{(n-v) \times (n-v)} \end{bmatrix} \quad (10)$$

To find a preimage for the central map $\mathcal{F}$, i.e. find $\mathbf{x} \in \mathbb{F}_q^n$ such that $\mathcal{F}(\mathbf{x}) = \mathbf{t}$ for a given $\mathbf{t} \in \mathbb{F}_q^m$, we try to randomly choose values $y_1, \dots, y_v \in \mathbb{F}_q$ as the values of the vinegar variables. Then it is easy to solve for the equation $\mathcal{F}(y_1, \dots, y_v, x_{v+1}, \dots, x_n) = \mathbf{t}$ because this is now a linear system of m equations in $n - v$ oil variables. For UOV scheme, we set $m = n - v$. If there is no solution then a new tuple of values $y'_1, \dots, y'_v$ is chosen and same process is repeated until success.

The process of signature for the Oil and Vinegar scheme is then clear: for given message $\mathbf{t} \in \mathbb{F}_q^m$ to be signed, first a solution $\mathbf{y} \in \mathbb{F}_q^n$ is derived from the equation $\mathcal{F}(\mathbf{y}) = \mathbf{t}$, then $\mathbf{s} := \mathcal{S}^{-1}(\mathbf{y})$ is the signature we want. The verification just checks if $\mathcal{P}(\mathbf{s}) = \mathbf{t}$.

Remark 2.5. 1. In the original paper [30], the author assumed that $n = 2m$ (i.e., $v = o = m$). However, this version was broken by Kipnis and Shamir in [26]. Therefore, it is necessary to have $v > o$, and the scheme is called the Unbalanced Oil and Vinegar (UOV) scheme.

2. Note that when $\text{char}(\mathbb{F}_q) \neq 2$, each $\mathbf{F}_k$ can be chosen symmetric. Hence p_k is represented by the symmetric matrix $\mathbf{S}^\top \cdot \mathbf{F}_k \cdot \mathbf{S}$. If $\text{char}(\mathbb{F}_q) = 2$, we represent p_k by the upper-triangular matrix $\text{Upper}(\mathbf{P}_k) \in \mathbf{M}_n(\mathbb{F}_q)$ with entries

$$\text{Upper}(\mathbf{P}_k)_{ij} = \begin{cases} (\mathbf{P}_k)_{ij} + (\mathbf{P}_k)_{ji}, & i < j, \\ (\mathbf{P}_k)_{ii}, & i = j, \\ 0, & i > j. \end{cases}$$

In §4.2, we will see that the size of the public key mainly depends on the dimension of $\mathcal{O}$, the number of equations m , and the field size q . The main idea of reducing the public key size in QR-UOV, MAYO, and SNOVA is to effectively decrease the dimension of the oil space $\mathcal{O}$ in the key generation phase.

3. In [6], Beullens utilized the relations in Eq. (5) satisfied by $\mathcal{P}$ to simplify the description of the UOV scheme and proposed an equivalent method, which uses the oil space, for finding a preimage of $\mathcal{P}$. The idea is as follows.

We first define the polar form of a multivariate quadratic polynomial $p(\mathbf{x})$ as

$$p'(\mathbf{x}, \mathbf{y}) := p(\mathbf{x} + \mathbf{y}) - p(\mathbf{x}) - p(\mathbf{y}) + p(0).$$

For the public quadratic map $\mathcal{P} = (p_1, \dots, p_m)$ in the UOV scheme, we define its polar form as

$$\mathcal{P}'(\mathbf{x}, \mathbf{y}) := (p'_1(\mathbf{x}, \mathbf{y}), \dots, p'_m(\mathbf{x}, \mathbf{y})).$$

Given a target $\mathbf{t} \in \mathbb{F}_q^m$, we first randomly choose $\mathbf{v} \in \mathbb{F}_q^n$ and solve the system $\mathcal{P}(\mathbf{v} + \mathbf{o}) = \mathbf{t}$ for a vector $\mathbf{o} \in \mathcal{O}$. This can be done by solving a linear system for $\mathbf{o}$:

$$\mathcal{P}(\mathbf{v} + \mathbf{o}) = \underbrace{\mathcal{P}(\mathbf{v})}_{\text{fixed by choice of } \mathbf{v}} + \underbrace{\mathcal{P}(\mathbf{o})}_{=0} + \underbrace{\mathcal{P}'(\mathbf{v}, \mathbf{o})}_{\text{linear function of } \mathbf{o}} = \mathbf{t}, \quad (11)$$

Since $\dim_{\mathbb{F}_q} \mathcal{O} = o$ and there are $m = o$ equations, we can solve this system with high probability. If the above system has no solution, one can simply pick a new $\mathbf{v}$ and try again. Then $\mathbf{s} = \mathbf{v} + \mathbf{o}$ is a valid signature for $\mathbf{t}$.

2.4.2 MAYO

MAYO[7] constructs smaller public key by whipping up a UOV public map $\mathcal{P}: \mathbb{F}_q^n \rightarrow \mathbb{F}_q^m$ over the base field $\mathbb{F}_q$ with small oil space into a k -fold larger map $\mathcal{P}^*: \mathbb{F}_q^{kn} \rightarrow \mathbb{F}_q^m$. The k -fold map is defined as

$$\mathcal{P}^*(\mathbf{x}_1, \dots, \mathbf{x}_k) := \sum_{i=1}^k \mathbf{E}_{i,i} \mathcal{P}(\mathbf{x}_i) + \sum_{i=1}^{k-1} \sum_{j=i+1}^k \mathbf{E}_{i,j} \mathcal{P}'(\mathbf{x}_i, \mathbf{x}_j) \quad (12)$$

where $\mathbf{x}_i = (x_{i,1}, \dots, x_{i,n})$ for $i \in [k]$, $\mathbf{E}_{i,j}$ are fixed public $m \times m$ matrices, and $\mathcal{P}'(x, y) := \mathcal{P}(\mathbf{x} + \mathbf{y}) - \mathcal{P}(\mathbf{x}) - \mathcal{P}(\mathbf{y})$ is the polar form for $\mathcal{P}$. The secret key is $(\mathcal{S}, \mathcal{F})$ and the public key is $(\mathcal{P}, \{\mathbf{E}_{i,j}\}_{i,j \in [k]})$.

Given a target $\mathbf{t} \in \mathbb{F}_q^m$, the signer computes a signature $\mathbf{s} = (\mathbf{s}_1, \dots, \mathbf{s}_k) \in \mathbb{F}_q^{kn}$ such that $\mathcal{P}^*(\mathbf{s}) = \mathbf{t}$. The signing process follows the method described in [4] (see also Remark 2.5 (2)) due to the fact that

$$\mathcal{P}^*(\mathbf{u}_1, \dots, \mathbf{u}_k) = 0 \quad \text{for any } \mathbf{u}_1, \dots, \mathbf{u}_k \in \mathcal{O}, \quad (13)$$

where $\mathcal{O}$ is the oil space of $\mathcal{P}$. The verification processes are the same as those for the plain UOV.

2.4.3 QR-UOV

QR-UOV[19] constructs smaller public key by using a subalgebra generated by one irreducible matrix (and its left multiplication shift) to form the public key. Given an irreducible polynomial over $\mathbb{F}_q$ of degree ℓ , we denote Φ_x to be its companion matrix. Define two subsets of the $\ell \times \ell$ matrix algebra $\mathfrak{R}$:

$$A_f = \{\lambda_0 \mathbf{I} + \lambda_1 \Phi_x + \lambda_2 \Phi_x^2 + \cdots + \lambda_{\ell-1} \Phi_x^{\ell-1} \mid \lambda_0, \dots, \lambda_{\ell-1} \in \mathbb{F}_q\}$$

$$\mathbf{W}A_f = \{\mathbf{W}\mathbf{X} \mid \mathbf{X} \in A_f\}$$

where $\mathbf{W}$ is an invertible and symmetric matrix chosen to make $\mathbf{W}A_f$ a subset of symmetric matrices. This makes sure that for $\mathbf{X} \in \text{Mat}_{n \times n}(\mathbb{F}_q) \otimes A_f$ and $\mathbf{Y} \in \text{Mat}_{n \times n}(\mathbb{F}_q) \otimes (\mathbf{W}A_f)$ we have

$$\mathbf{X}^\top \mathbf{Y} \mathbf{X} \in \text{Mat}_{n \times n}(\mathbb{F}_q) \otimes (\mathbf{W}A_f).$$

[19] proved that such $\mathbf{W}$ always exists. QR-UOV then chooses $\mathbf{F}_k$ for $k \in [m]$ from $\text{Mat}_{n \times n}(\mathbb{F}_q) \otimes (\mathbf{W}A_f)$ with lower-right $(n-v) \times (n-v)$ block zero as in Eq. (6), chooses invertible $\mathbf{S}$ from $\text{Mat}_{n \times n}(\mathbb{F}_q) \otimes A_f$, and compute $\mathbf{P}_i$ as in Eq. (9). The secret key is $(\mathbf{S}, \{\mathbf{F}_k\}_{k \in [m]})$ and the public key is $\{\mathbf{P}_k := \mathbf{S}^\top \mathbf{F}_k \mathbf{S}\}_{k \in [m]}$.

2.4.4 SNOVA

SNOVA[39] constructs smaller public key by shifting base ring directly to the algebra $\mathfrak{R}$ of $\ell \times \ell$ matrices. Given a symmetric $\ell \times \ell$ matrix $\mathbf{S}$ with irreducible characteristic polynomial, the scheme uses the subalgebra

$$\mathbb{F}_q[\mathbf{S}] = \{\lambda_0 \mathbf{I} + \lambda_1 \mathbf{S} + \lambda_2 \mathbf{S}^2 + \cdots + \lambda_{\ell-1} \mathbf{S}^{\ell-1} \mid \lambda_0, \dots, \lambda_{\ell-1} \in \mathbb{F}_q\}$$

Let

$$\Omega = \{(i, j) \mid 1 \leq i, j \leq n\} \setminus \{(i, j) \mid v+1 \leq i, j \leq n\}$$

The central map $f_k: \mathfrak{R}^n \rightarrow \mathfrak{R}^m$ is defined as

$$\begin{aligned} f_k(\mathbf{X}_1, \dots, \mathbf{X}_n) &= \sum_{\alpha=0}^{\ell^2+\ell-1} \mathbf{A}_{k,\alpha} \left(\sum_{(i,j) \in \Omega} \mathbf{X}_i^\top \mathbf{Q}_{k,\alpha,1} \mathbf{F}_{k',i,j} \mathbf{Q}_{k,\alpha,2} \mathbf{X}_j \right) \mathbf{B}_{k,\alpha} \\ &= \sum_{\alpha=0}^{\ell^2+\ell-1} \mathbf{A}_{k,\alpha} \mathbf{U}^\top \Lambda_{\mathbf{Q}_{k,\alpha,1}}^{(n)} \mathbf{F}_{k'} \Lambda_{\mathbf{Q}_{k,\alpha,2}}^{(n)} \mathbf{U} \mathbf{B}_{k,\alpha} \end{aligned} \quad (14)$$

where $k' = k + \alpha \pmod m$, $\mathbf{X}_i$ are $\ell \times \ell$ matrix variables, $\mathbf{U} = [\mathbf{X}_1^\top \ \dots \ \mathbf{X}_n^\top]^\top$, $\mathbf{A}_{k,\alpha}$ and $\mathbf{B}_{k,\alpha}$ are randomly chosen invertible elements in $\mathfrak{R}$, $\mathbf{Q}_{k,\alpha,1}$ and $\mathbf{Q}_{k,\alpha,2}$ are randomly chosen invertible elements in $\mathbb{F}_q[\mathbf{S}]$, $\mathbf{F}_{k',i,j}$ are randomly chosen elements in $\mathfrak{R}$ and $\mathbf{F}_{k'}$ is the $n \times n$ block matrix with $\mathbf{F}_{k',i,j}$ as the (i, j) -th block.

The invertible linear map $\mathcal{T}: \mathfrak{R}^n \rightarrow \mathfrak{R}^n$ and corresponding $\mathbf{T}$ is chosen as in Eq. (10) with the difference that all the entries should be viewed as in $\mathfrak{R}$ and the elements in the upper-right part are all in $\mathbb{F}_q[\mathbf{S}]$. This makes sure that

$$\mathbf{T}^\top \Lambda_{\mathbf{Q}_{k,\alpha,1}}^{(n)} \mathbf{F}_{k'} \Lambda_{\mathbf{Q}_{k,\alpha,2}}^{(n)} \mathbf{T} = \Lambda_{\mathbf{Q}_{k,\alpha,1}}^{(n)} \mathbf{T}^\top \mathbf{F}_{k'} \mathbf{T} \Lambda_{\mathbf{Q}_{k,\alpha,2}}^{(n)}$$

and gives us

$$p_k(\mathbf{X}_1, \dots, \mathbf{X}_n) = \sum_{\alpha=0}^{\ell^2+\ell-1} \mathbf{A}_{k,\alpha} \mathbf{U}^\top \Lambda_{\mathbf{Q}_{k,\alpha,1}}^{(n)} \mathbf{P}_{k'} \Lambda_{\mathbf{Q}_{k,\alpha,2}}^{(n)} \mathbf{U} \mathbf{B}_{k,\alpha} \quad (15)$$

where $\mathbf{P}_{k'} = \mathbf{T}^\top \mathbf{F}_{k'} \mathbf{T}$ and $k' = k + \alpha \pmod{m}$. The secret key is $(\mathbf{T}, \{\mathbf{F}_i\}_{i \in [m]})$ and the public key is

$$\{\mathbf{A}_{i,\alpha}, \mathbf{B}_{i,\alpha}, \mathbf{Q}_{i,\alpha,1}, \mathbf{Q}_{i,\alpha,2}, \mathbf{P}_i : i \in [m], 0 \leq \alpha \leq \ell^2 + \ell - 1\}.$$

Since SNOVA can be regarded as a UOV scheme over $\mathbb{F}_q$, the signing and verification procedures are the same as those for the plain UOV. See [40] for more details.

3 Design rationale

3.1 Tensor structure of UOV variants

The structure of using matrix algebra's elements as entries for calculation encourages us to dig into the tensor product language and give these UOV variants some new expressions.

3.1.1 MAYO

With the same notations as in §2.4.2. Suppose that $\mathcal{P}^* = (p_1^*, \dots, p_m^*)$ and

$$p_s^*(\mathbf{x}_1, \dots, \mathbf{x}_k) = (\mathbf{x}_1^\top, \dots, \mathbf{x}_k^\top) \cdot \mathbf{P}_s^* \cdot (\mathbf{x}_1^\top, \dots, \mathbf{x}_k^\top)^\top, \quad \mathbf{P}_s^* \in \text{Mat}_{kn \times kn}(\mathbb{F}_q), s \in [m].$$

In the following, we will give a tensor representation for $\mathbf{P}_s^*$.

Let $\mathbf{P}_i$ be a matrix such that $p_i(\mathbf{x}) = \mathbf{x}^\top \cdot \mathbf{P}_i \cdot \mathbf{x}$ for $i \in [m]$. Note that

$$p'_i(\mathbf{x}, \mathbf{y}) := p_i(\mathbf{x} + \mathbf{y}) - p_i(\mathbf{x}) - p_i(\mathbf{y}) + p_i(0) = \mathbf{x}^\top \cdot \mathbf{P}_i \cdot \mathbf{y} + \mathbf{y}^\top \cdot \mathbf{P}_i \cdot \mathbf{x}.$$

Denote the (a, b) -th entry of E_{ij} by e_{ij}^{ab} . By Eq. (12), the public key p_s^* can be rewritten

as follows:

$$\begin{aligned}
p_s^*(\mathbf{x}_1, \dots, \mathbf{x}_k) &= \sum_{i=1}^k \sum_{a=1}^m e_{ii}^{sa} p_a(\mathbf{x}_i) + \sum_{i=1}^k \sum_{j=i+1}^k \sum_{a=1}^m e_{ij}^{sa} p'_a(\mathbf{x}_i, \mathbf{x}_j) \\
&= \sum_{i=1}^k \sum_{a=1}^m e_{ii}^{sa} \mathbf{x}_i^\top \mathbf{P}_a \mathbf{x}_i + \sum_{1 \leq i < j \leq k} \sum_{a=1}^m e_{ij}^{sa} (\mathbf{x}_i^\top \mathbf{P}_a \mathbf{x}_j + \mathbf{x}_j^\top \mathbf{P}_a \mathbf{x}_i) \\
&= \sum_{i=1}^k \mathbf{x}_i^\top \left(\sum_{a=1}^m e_{ii}^{sa} \mathbf{P}_a \right) \mathbf{x}_i + \sum_{1 \leq i < j \leq k} \left(\mathbf{x}_i^\top \cdot \left(\sum_{a=1}^m e_{ij}^{sa} \mathbf{P}_a \right) \cdot \mathbf{x}_j \right. \\
&\quad \left. + \mathbf{x}_j^\top \cdot \left(\sum_{a=1}^m e_{ij}^{sa} \mathbf{P}_a \right) \cdot \mathbf{x}_i \right) \\
&= (\mathbf{x}_1^\top, \dots, \mathbf{x}_k^\top) \cdot \begin{pmatrix} \sum_{a=1}^m e_{11}^{sa} \mathbf{P}_a & \sum_{a=1}^m e_{12}^{sa} \mathbf{P}_a & \cdots \\ \sum_{a=1}^m e_{12}^{sa} \mathbf{P}_a & \sum_{a=1}^m e_{22}^{sa} \mathbf{P}_a & \cdots \\ \vdots & \vdots & \ddots \end{pmatrix} \cdot (\mathbf{x}_1^\top, \dots, \mathbf{x}_k^\top)^\top \\
&= (\mathbf{x}_1^\top, \dots, \mathbf{x}_k^\top) \cdot \mathbf{P}_s^* \cdot (\mathbf{x}_1^\top, \dots, \mathbf{x}_k^\top)^\top,
\end{aligned} \tag{16}$$

where

$$\mathbf{P}_s^* = \sum_{a=1}^m \mathbf{E}^{sa} \otimes \mathbf{P}_a, \text{ and the } (i, j)\text{-th entry of } \mathbf{E}^{sa} = \begin{cases} e_{ij}^{sa} & \text{if } i \leq j, \\ e_{ji}^{sa} & \text{if } i > j. \end{cases} \tag{17}$$

Let $\mathcal{O}$ be the oil space of $\mathcal{P}$ in MAYO, then for any

$$\mathbf{v} \in \mathbb{F}_q^k \otimes \mathcal{O} = \left\{ \sum_{i=1}^s \alpha_i \otimes \mathbf{u}_i : \alpha_i \in \mathbb{F}_q^k, \mathbf{u}_i \in \mathcal{O} \right\} \subset \mathbb{F}_q^{kn}, \tag{18}$$

we have $\mathbf{v}^\top \cdot \mathbf{P}_s^* \cdot \mathbf{v} = 0$. Note that $\dim(\mathbb{F}_q^k \otimes \mathcal{O}) = ko$, which explains how MAYO enlarges the oil dimension from o to ko in the view of tensor language.

We collect same terms in the coordinates of $\mathcal{P}$. Denote $e_{i,j}^{(r,s)}$ to be the (r, s) -th element of $\mathbf{E}_{i,j}$ when $i \leq j$ and extend this definition to $i > j$ by making the definition symmetry in i and j . We have

$$\begin{aligned}
\tilde{p}_r(\mathbf{x}_1, \dots, \mathbf{x}_k) &= \sum_{i=1}^k \sum_{s=1}^m e_{i,i}^{(r,s)} p_s(\mathbf{x}_i) + \sum_{i=1}^{k-1} \sum_{j=i+1}^k \sum_{s=1}^m e_{i,j}^{(r,s)} p'_s(\mathbf{x}_i, \mathbf{x}_j) \\
&= \sum_{i=1}^k \sum_{s=1}^m e_{i,i}^{(r,s)} \mathbf{x}_i^\top \mathbf{P}_s \mathbf{x}_i + \sum_{i=1}^{k-1} \sum_{j=i+1}^k \sum_{s=1}^m e_{i,j}^{(r,s)} (\mathbf{x}_i^\top \mathbf{P}_s \mathbf{x}_j + \mathbf{x}_j^\top \mathbf{P}_s \mathbf{x}_i) \\
&= [\mathbf{x}_1^\top \quad \dots \quad \mathbf{x}_k^\top] \left(\sum_{s=1}^m (e_{i,j}^{(r,s)})_{i,j} \otimes \mathbf{P}_s \right) \begin{bmatrix} \mathbf{x}_1 \\ \vdots \\ \mathbf{x}_k \end{bmatrix}
\end{aligned}$$

Therefore if we denote $\tilde{\mathbf{E}}^{(r,s)}$ to be the matrix whose (i, j) -th element is $e_{i,j}^{(r,s)}$, then we have

$$\tilde{\mathbf{P}}_r = \sum_{s=1}^m \tilde{\mathbf{E}}^{(r,s)} \otimes \mathbf{P}_s \quad (19)$$

3.1.2 QR-UOV

By definition of A_f and $\mathbf{W}A_f$ we know there exists $\mathbf{F}_{k,d} \in \text{Mat}_{n \times n}(\mathbb{F}_q)$ for $k \in [m]$ and $d \in [\ell]$, such that

$$\mathbf{F}_k = \sum_{d=1}^{\ell} \mathbf{F}_{k,d} \otimes (\mathbf{W}\Phi_x^{d-1}) \quad (20)$$

Similarly, there exists $\mathbf{S}'_d \in \text{Mat}_{v \times (n-v)}(\mathbb{F}_q)$ for $d \in [\ell]$ such that

$$\begin{aligned} \mathbf{S} &= \begin{bmatrix} \mathbf{I}_{vl \times vl} & \sum_{d=1}^{\ell} \mathbf{S}'_d \otimes \Phi_x^{d-1} \\ \mathbf{0}_{(nl-vl) \times vl} & \mathbf{I}_{(nl-vl) \times (nl-vl)} \end{bmatrix} \\ &= \sum_{d=1}^{\ell} \mathbf{S}_d \otimes \Phi_x^{d-1}, \end{aligned} \quad (21)$$

where

$$\mathbf{S}_1 = \begin{bmatrix} \mathbf{I}_{v \times v} & \mathbf{S}'_1 \\ \mathbf{0}_{(n-v) \times v} & \mathbf{I}_{(n-v) \times (n-v)} \end{bmatrix}, \quad \mathbf{S}_d = \begin{bmatrix} \mathbf{0}_{v \times v} & \mathbf{S}'_d \\ \mathbf{0}_{(n-v) \times v} & \mathbf{0}_{(n-v) \times (n-v)} \end{bmatrix}, \quad d = 2, \dots, \ell.$$

3.1.3 SNOVA

Let $p_{i,c,d}^*(\mathbf{U})$ be the (c, d) -th entry of $p_i^*(\mathbf{U})$ and suppose that

$$p_{i,c,d}^*(\mathbf{U}) = \text{vec}(\mathbf{U})^t \cdot \mathbf{P}_{i,c,d}^* \cdot \text{vec}(\mathbf{U}), \quad \mathbf{P}_{i,c,d}^* \in \text{Mat}_{n\ell^2 \times n\ell^2}(\mathbb{F}_q). \quad (22)$$

In the following, we will give a tensor representation for $\mathbf{P}_{i,c,d}^*$.

Suppose that

$$\Lambda_{\mathbf{Q}_{i,\alpha,1}}^{(n)} \cdot \mathbf{P}_{i'} \cdot \Lambda_{\mathbf{Q}_{i,\alpha,2}}^{(n)} = \sum_{0 \leq a,b \leq \ell-1} c_{i,\alpha}^{a,b} \cdot \Lambda_{\mathbf{S}^a}^{(n)} \cdot \mathbf{P}_{i'} \cdot \Lambda_{\mathbf{S}^b}^{(n)} \quad \text{where } c_{i,\alpha}^{a,b} \in \mathbb{F}_q$$

for $i, i' \in [o]$ and define $\mathbf{P}_{i,a,b} := \Lambda_{\mathbf{S}^a}^{(n)} \cdot \mathbf{P}_i \cdot \Lambda_{\mathbf{S}^b}^{(n)}$. Then, we have

$$\begin{aligned} \text{vec}(p_i^*(\mathbf{U})) &= \sum_{\alpha=0}^{\ell^2+\ell-1} \mathbf{B}_{i,\alpha}^\top \otimes \mathbf{A}_{i,\alpha} \cdot \text{vec}(\mathbf{U}^\top \cdot \Lambda_{\mathbf{Q}_{i,\alpha,1}}^{(n)} \cdot \mathbf{P}_{i'} \cdot \Lambda_{\mathbf{Q}_{i,\alpha,2}}^{(n)} \cdot \mathbf{U}) \\ &= \sum_{\alpha=0}^{\ell^2+\ell-1} \sum_{0 \leq a,b \leq \ell-1} c_{i,\alpha}^{a,b} \cdot (\mathbf{B}_{i,\alpha}^\top \otimes \mathbf{A}_{i,\alpha}) \cdot \text{vec}(\mathbf{U}^\top \cdot \mathbf{P}_{i',a,b} \cdot \mathbf{U}) \\ &= \sum_{\alpha=0}^{\ell^2+\ell-1} \sum_{0 \leq a,b \leq \ell-1} \mathbf{T}_{i,\alpha}^{a,b} \cdot \text{vec}(\mathbf{U}^\top \cdot \mathbf{P}_{i',a,b} \cdot \mathbf{U}), \end{aligned} \quad (23)$$

where $\mathbf{T}_{i,\alpha}^{a,b} := c_{i,\alpha}^{a,b} \cdot (\mathbf{B}_{i,\alpha}^\top \otimes \mathbf{A}_{i,\alpha})$. Thus, we have

$$\text{vec}(p_i^*(\mathbf{U}))_r = \sum_{\alpha=0}^{\ell^2+\ell-1} \sum_{0 \leq a,b \leq \ell-1} \sum_{s=1}^{\ell^2} \mathbf{T}_{i,\alpha}^{a,b}(r,s) \cdot \text{vec}(\mathbf{U}^\top \cdot \mathbf{P}_{i',a,b} \cdot \mathbf{U})_s, \quad (24)$$

where $\text{vec}(\cdot)_r$ denotes the r -th component of a vector and $\mathbf{T}_{i,\alpha}^{a,b}(r,s)$ the (r,s) -entry of $\mathbf{T}_{i,\alpha}^{a,b}$.

Note that the (e,f) -th entry of $\mathbf{U}^t \cdot \mathbf{P}_{i',a,b} \cdot \mathbf{U}$ is the $e + (f-1)\ell$ -th component of $\text{vec}(\mathbf{U}^t \cdot \mathbf{P}_{i',a,b} \cdot \mathbf{U})$, which can also be expressed as

$$(\mathbf{U}^t \cdot \mathbf{P}_{i',a,b} \cdot \mathbf{U})(e,f) = \text{vec}(\mathbf{U})^t \cdot (I_{e,f} \otimes \mathbf{P}_{i',a,b}) \cdot \text{vec}(\mathbf{U}), \quad (25)$$

where $I_{e,f}$ is the matrix with 1 in the (e,f) -th position and zeros elsewhere. Combining with Eq. (24) and Eq. (25), we have

$$\mathbf{P}_{i,c,d}^* = \sum_{\alpha=0}^{\ell^2+\ell-1} \sum_{0 \leq a,b \leq \ell-1} \sum_{e,f \in [\ell]} \mathbf{T}_{i,\alpha}^{a,b}(c + (d-1)\ell, e + (f-1)\ell) \cdot I_{e,f} \otimes \mathbf{P}_{i',a,b}. \quad (26)$$

Define

$$\mathbf{E}_{i,\alpha}^{a,b,c,d} := \sum_{e,f \in [\ell]} \mathbf{T}_{i,\alpha}^{a,b}(c + (d-1)\ell, e + (f-1)\ell) \cdot I_{e,f} \in \mathbf{M}_\ell(\mathbb{F}_q).$$

We have

$$\mathbf{P}_{i,c,d}^* = \sum_{\alpha=0}^{\ell^2+\ell-1} \sum_{0 \leq a,b \leq \ell-1} \mathbf{E}_{i,\alpha}^{a,b,c,d} \otimes \mathbf{P}_{i',a,b}, \quad (27)$$

where $i' = i + \alpha \pmod m$.

4 Algorithm Specification

4.1 Algorithm Description

In this subsection, we present the detailed specifications of the proposed signature scheme, TSUOV. The design of TSUOV is inspired by the structure of QR-UOV, MAYO, and SNOVA, and can be regarded as a unification and generalization of these schemes.

Set up.

- Take integers n, o, ℓ, k, m_1, m_2 and q a power of some prime number such that $n > o$ and $olk \geq m_2 \geq m_1$. Set $v = n - o$.
- Take a symmetric matrix $\Phi \in \text{Mat}_{\ell \times \ell}(\mathbb{F}_q)$ whose characteristic polynomial is irreducible. The existence of such a matrix is established in [2, Corollary 4.2] for even characteristic and in [27] for odd characteristic. Fix an isomorphism $\mathbb{F}_q[\Phi] \cong \mathbb{F}_{q^\ell}$, denote the image of Φ under this isomorphism by $\lambda \in \mathbb{F}_q^\ell$.

Key Generation.

- Choose randomly matrices $\bar{\mathbf{F}}_i \in \text{Mat}_{n \times n}(\mathbb{F}_{q^\ell})$ ($i \in [m_1]$) such that the lower right $o \times o$ submatrix are zero matrices. Choose randomly matrix $\bar{\mathbf{S}}' \in \text{Mat}_{v \times o}(\mathbb{F}_{q^\ell})$. Suppose that

$$\bar{\mathbf{S}} = \begin{pmatrix} \mathbf{I}_v & \bar{\mathbf{S}}' \\ 0 & \mathbf{I}_o \end{pmatrix},$$

where $\mathbf{I}_v$ and $\mathbf{I}_o$ are identity matrices of size $v \times v$ and $o \times o$, respectively. Suppose that $\bar{\mathbf{P}}_i = \bar{\mathbf{S}}^\top \cdot \bar{\mathbf{F}}_i \cdot \bar{\mathbf{S}}$ for $i \in [m_1]$.

- We can uniformly write each $\bar{\mathbf{F}}_i$, $\bar{\mathbf{S}}$, and $\bar{\mathbf{P}}_i$ as follows and define

$$\begin{aligned} \bar{\mathbf{F}}_i &= \sum_{j=0}^{\ell-1} \mathbf{F}_{ij} \cdot \lambda^j, \quad \mathbf{F}_{ij} \in \text{Mat}_{n \times n}(\mathbb{F}_q), \quad \mathbf{F}_i := \sum_{j=0}^{\ell-1} \mathbf{F}_{ij} \otimes \Phi^j \in \text{Mat}_{n\ell \times n\ell}(\mathbb{F}_q), \\ \bar{\mathbf{S}} &= \sum_{j=0}^{\ell-1} \mathbf{S}_j \cdot \lambda^j, \quad \mathbf{S}_j \in \text{Mat}_{n \times n}(\mathbb{F}_q), \quad \mathbf{S} := \sum_{j=0}^{\ell-1} \mathbf{S}_j \otimes \Phi^j \in \text{Mat}_{n\ell \times n\ell}(\mathbb{F}_q), \\ \bar{\mathbf{P}}_i &= \sum_{j=0}^{\ell-1} \mathbf{P}_{ij} \cdot \lambda^j, \quad \mathbf{P}_{ij} \in \text{Mat}_{n \times n}(\mathbb{F}_q), \quad \mathbf{P}_i := \sum_{j=0}^{\ell-1} \mathbf{P}_{ij} \otimes \Phi^j \in \text{Mat}_{n\ell \times n\ell}(\mathbb{F}_q). \end{aligned} \tag{28}$$

One can verify that $\mathbf{P}_i = \mathbf{S}^\top \cdot \mathbf{F}_i \cdot \mathbf{S}$ for $i \in [m_1]$ easily.

- Take matrices $\mathbf{E}^{sa} \in \text{Mat}_{k \times k}(\mathbb{F}_q)$ randomly for $s \in [m_2], a \in [m_1]$. Define

$$\begin{aligned} \mathbf{F}_s^* &= \sum_{a=1}^{m_1} \mathbf{E}^{sa} \otimes \mathbf{F}_a, \quad f_s^*(\mathbf{x}) = \mathbf{x}^\top \mathbf{F}_s^* \mathbf{x}, \quad s \in [m_2], \\ \mathbf{P}_s^* &= \sum_{a=1}^{m_1} \mathbf{E}^{sa} \otimes \mathbf{P}_a, \quad p_s^*(\mathbf{x}) = \mathbf{x}^\top \mathbf{P}_s^* \mathbf{x}, \quad s \in [m_2]. \end{aligned} \tag{29}$$

The central map is $\mathcal{F}^* = (f_1^*, \dots, f_{m_2}^*) : \mathbb{F}_q^{n\ell k} \rightarrow \mathbb{F}_q^{m_2}$ and the public map is $\mathcal{P}^* = (p_1^*, \dots, p_{m_2}^*) : \mathbb{F}_q^{n\ell k} \rightarrow \mathbb{F}_q^{m_2}$. One can check that $\mathcal{P}^* = \mathcal{F}^* \circ \mathcal{S}$, where $\mathcal{S} : \mathbb{F}_q^{n\ell k} \rightarrow \mathbb{F}_q^{n\ell k}$ is the linear map corresponding to the matrix $\mathbf{I}_k \otimes \mathbf{S}$. In particular, we have

$$\mathbf{P}_s^* = (\mathbf{I}_k \otimes \mathbf{S}^\top) \cdot \mathbf{F}_s^* \cdot (\mathbf{I}_k \otimes \mathbf{S}) \quad \text{for } s \in [m_2]. \tag{30}$$

- The secret key is $(\mathbf{S}, \{\mathbf{F}_i\}_{i \in [m_1]})$, and the public key consists of $\{\mathbf{P}_i\}_{i \in [m_1]}$ together with $\{\mathbf{E}^{sa}\}_{s \in [m_2], a \in [m_1]}$. Equivalently, for each $i \in [m_1]$, one may work with $(\bar{\mathbf{S}}, \bar{\mathbf{F}}_i, \bar{\mathbf{P}}_i)$ in place of $(\mathbf{S}, \mathbf{F}_i, \mathbf{P}_i)$.

Signing.

- Let $\mathbf{msg} \in \{0, 1\}^*$ be a message to be signed. Take a cryptographic hash function $\mathbf{Hash} : \{0, 1\}^* \rightarrow \mathbb{F}_q^{m_2}$ and $\mathbf{salt} \in \{0, 1\}^\lambda$. Suppose that $\mathbf{t} = \mathbf{Hash}(\mathbf{msg} || \mathbf{salt})$.

- To generate a signature $\mathbf{s} \in \mathbb{F}_q^{n\ell k}$ for a target $\mathbf{t}$, we invert the private map $\mathcal{F}^*$ and then apply $\mathcal{S}^{-1}$. First, sample the vinegar variables $a_{1,1}, \dots, a_{k,v\ell} \in \mathbb{F}_q$ uniformly at random and plug them into

$$\mathcal{F}^*(\mathbf{x}) = \mathbf{t}, \quad \mathbf{x} = (\mathbf{x}_1, \dots, \mathbf{x}_k)^\top \quad (31)$$

where $\mathbf{x}_i = (a_{i,1}, \dots, a_{i,v\ell}, x_{i,1}, \dots, x_{i,o\ell})$. This yields a linear system of m_2 equations in the olk oil variables $\{x_{i,j}\}$, which we solve to obtain $\mathbf{x}$. If the system has no solution, resample the vinegar variables and repeat. Finally, output the signature $(\mathbf{s} = \mathcal{S}^{-1}(\mathbf{x}), \text{salt})$.

Verification.

Let $(\mathbf{s} \in \mathbb{F}_q^{n\ell k}, \text{salt})$ be the signature to be verified. If $\text{Hash}(\text{msg}||\text{salt}) = \mathcal{P}^*(\mathbf{s})$, then the signature is accepted, otherwise rejected.

- Remark 4.1.*
1. Compared with QR-UOV, MAYO, and SNOVA, the matrices $\mathbf{P}_i^*$ in TSUOV involve two tensor products, which enlarge the oil space from dimension o to olk . Moreover, we explicitly distinguish the number of equations used in key generation, m_1 , from that used in signing, m_2 . These modifications enlarge the design space and enable a more flexible trade-off between security and efficiency.
 2. In TSUOV, the key-generation procedure differs slightly depending on the characteristic of the base field. If $\text{char}(\mathbb{F}_q)$ is odd, we may assume that each $\mathbf{F}_i$ is symmetric. If $\text{char}(\mathbb{F}_q) = 2$, we replace $\mathbf{P}_i$ by $\text{Upper}(\mathbf{P}_i)$ and assume that each $\mathbf{E}^{sa}$ is symmetric. Since $\mathbf{P}_i$ can be viewed as a matrix over the algebra $\mathbb{F}_q[\Phi]$ and Φ is symmetric, when $\ell > 1$, each $\ell \times \ell$ diagonal block of $\text{Upper}(\mathbf{P}_i)$ is itself diagonal. Consequently, when defining the quadratic polynomials via $\mathbf{P}_i^*$, certain quadratic terms vanish automatically, which complicates the choice of secure parameters. For this reason, when $\text{char}(\mathbb{F}_q) = 2$ we recommend restricting to $\ell = 1$. In the examples reported at the end of this paper, we focus on parameter sets over fields of odd characteristic.
 3. In Eq. (28), if the characteristic polynomial of Φ is reducible, then each $\mathbf{F}_i$ and $\mathbf{P}_i$ can be block-diagonalized. This leads to vulnerabilities similar to those in BAC-UOV[20]. Hence, we insist that the characteristic polynomial of Φ is irreducible.

Remark 4.2. The proposed scheme can be regarded as a unification and generalization of QR-UOV, MAYO, and SNOVA. Compared with ad-hoc polynomial formulations, the tensor formalism not only simplifies notation but also reveals deeper connections among existing variants.

1. When $\ell = 1, k = 1$ and $m_1 = m_2 = m = o$, the above scheme is the original UOV.
2. When $k = 1, m_1 = m_2 = m$ and $\text{char}(\mathbb{F}_q)$ is odd, each $\bar{\mathbf{F}}_i$ can be chosen to be symmetric, and our construction essentially reduces to QR-UOV. A key technical difference concerns how we ensure that

$$\bar{\mathbf{P}}_i = \bar{\mathbf{S}}^\top \cdot \bar{\mathbf{F}}_i \cdot \bar{\mathbf{S}} \Rightarrow \mathbf{P}_i = \mathbf{S}^\top \cdot \mathbf{F}_i \cdot \mathbf{S}. \quad (32)$$

In QR-UOV, the block representation is generated by the companion matrix Φ_x of an irreducible polynomial f of degree ℓ , and matrices are taken from the $\mathbb{F}_q$ -algebra $A_f (= \mathbb{F}_q[\Phi_x])$. However, A_f is generally not closed under transpose: for $\Phi_h, \Phi_g \in A_f$, the product $(\Phi_h)^\top \cdot \Phi_g$ need not belong to A_f . To restore a symmetric form within A_f , QR-UOV introduces an auxiliary matrix $\mathbf{W}$ as a symmetrizer, thus preserve the relation Eq. (32) above over $\mathbb{F}_q$. In contrast, our scheme works with symmetric matrices directly, so the extra matrix $\mathbf{W}$ is unnecessary, leading to a simpler and more concise construction.

3. When $\ell = 1, m_1 = m_2 = m$ and each $\mathbf{E}^{sa}$ is symmetric, the above scheme is MAYO. Indeed, since the entries of $\{\mathbf{E}^{sa}\}$ are essentially a rearrangement of the entries of $\{\mathbf{E}_{ij}\}$ in MAYO, the whipping structure of MAYO is equivalent to the language of tensors. If we rewrite our scheme in the whipping structure, the corresponding matrices $\mathbf{E}_{ij}$ become rectangular of size $m_2 \times m_1$, whereas in MAYO they are square of size $m \times m$. This extra flexibility is one reason why our public key can be smaller than MAYO's.

However, the tensor-based framework provides a unified language to describe QR-UOV and MAYO. Therefore, we believe that the tensor formalism facilitates the construction of more general signature schemes and offers greater flexibility in parameter selection.

4. When $\ell = 1, m_1 = m_2 = o\ell^2$, and the public matrices are chosen as $\Lambda_{\mathbf{S}^a}^{(n)} \cdot \mathbf{P}_i \cdot \Lambda_{\mathbf{S}^b}^{(n)}$, the above scheme is SNOVA. The idea behind TSUOV is somewhat similar to that of SNOVA. In their scheme, the key generation step only requires generating o public keys $\mathbf{P}_1, \dots, \mathbf{P}_o$, but signing relies on utilizing $o\ell^2$ equations. Additionally, the direct use of symmetric matrices in place of the companion matrices in QR-UOV is also inspired by SNOVA.

Remark 4.3. As explained above, the tensor formulation and the whipping formulation are equivalent. In MAYO, the matrices $\mathbf{E}_{ij}$ are square and are chosen so that every nonzero linear combination is invertible. In our setting, $\mathbf{E}_{ij}$ is rectangular, so invertibility is replaced by the requirement of full column rank. Concretely, for each $i, j \in [k]$, define $\mathbf{E}_{ij} \in \mathbb{F}_q^{m_2 \times m_1}$ by

$$(\mathbf{E}_{ij})_{s,a} := (\mathbf{E}^{sa})_{i,j}, \quad s \in [m_2], a \in [m_1].$$

Hence, fixing $\{\mathbf{E}_{ij}\}$ is equivalent to fixing $\{\mathbf{E}^{sa}\}$.

To instantiate these matrices, let $g(z) \in \mathbb{F}_q[z]$ be irreducible of degree m_2 , and work in the quotient ring $\mathbb{F}_q[z]/\langle g(z) \rangle$. For each i, j , take $\mathbf{E}_{ij}$ to be the submatrix formed by the first m_1 columns of the matrix representation of multiplication by $z^{j+i(k-1)}$. With this choice, every nonzero linear combination of $\mathbf{E}_{ij}$ has full column rank. This design is motivated by security considerations; see Section 6.3.1.

4.2 Representation of Keys and Signature

Due to the structure of the new scheme, we can use the data $\bar{\mathbf{P}}_i, \bar{\mathbf{S}}, \bar{\mathbf{F}}_i$ to recover the P_i, S, F_i . Thus we only need to consider the data $\bar{\mathbf{P}}_i, \bar{\mathbf{S}}, \bar{\mathbf{F}}_i$. Based on Petzoldt et al.'s

compression technique [32], we can replace a large part of the public key with a small seed for pseudo-random number generation. In the following, we suppose that $\text{char}(\mathbb{F}_q)$ is odd. In fact, this method can also be applied to the even characteristic case. Then we may assume that $\mathbf{P}_i, \mathbf{F}_i$ are symmetric matrices with the following block forms:

$$\bar{\mathbf{P}}_i = \begin{pmatrix} \bar{\mathbf{P}}_{i,1} & \bar{\mathbf{P}}_{i,2} \\ \bar{\mathbf{P}}_{i,2}^\top & \bar{\mathbf{P}}_{i,3} \end{pmatrix}, \quad \bar{\mathbf{F}}_i = \begin{pmatrix} \bar{\mathbf{F}}_{i,1} & \bar{\mathbf{F}}_{i,2} \\ \bar{\mathbf{F}}_{i,2}^\top & 0_{o \times o} \end{pmatrix}, \quad \text{and } \bar{\mathbf{S}} = \begin{pmatrix} \mathbf{I}_{v \times v} & \mathbf{S}' \\ 0_{o \times v} & \mathbf{I}_{o \times o} \end{pmatrix} \quad (33)$$

From the equality $\bar{\mathbf{P}}_i = \bar{\mathbf{S}}^\top \cdot \bar{\mathbf{F}}_i \cdot \bar{\mathbf{S}}$, we have

$$\begin{aligned} \bar{\mathbf{F}}_{i,1} &= \bar{\mathbf{P}}_{i,1} \\ \bar{\mathbf{F}}_{i,2} &= -\bar{\mathbf{P}}_{i,1} \mathbf{S}' + \bar{\mathbf{P}}_{i,2} \\ \bar{\mathbf{P}}_{i,3} &= -\mathbf{S}'^\top \bar{\mathbf{P}}_{i,1} \mathbf{S}' + \mathbf{S}'^\top \bar{\mathbf{P}}_{i,2} + \bar{\mathbf{P}}_{i,2}^\top \mathbf{S}'. \end{aligned} \quad (34)$$

Thus in the key generation step, $\bar{\mathbf{P}}_{i,1}, \bar{\mathbf{P}}_{i,2}$ and $\mathbf{S}'$ can be generated first from the seeds seed_{pk} and seed_{sk} . Next, $\bar{\mathbf{P}}_{i,3}$ is computed using the last equation in Eq. (34). Furthermore, the small matrices $\mathbf{E}^{sa}$ can also be generated by seed_{pk} . Therefore, the public key is composed of $o \times o$ matrices $\bar{\mathbf{P}}_{i,3}$ for $i \in [m_1]$ and the seed seed_{pk} for $\bar{\mathbf{P}}_{i,1}, \bar{\mathbf{P}}_{i,2} (i \in [m_1])$ and $\mathbf{E}^{sa}$. In the following, we describe the representations of the public and private keys and the signature.

- The public key pk is a concatenation of byte strings representing $\text{seed}_{\text{pk}} \in \{0, 1\}^\lambda$ and $\bar{\mathbf{P}}_{i,3} (i \in [m_1])$, where λ is a security parameter. Therefore, when we ignore Φ , the approximate size of the public key is

$$(\lceil \log_2 q \rceil \cdot m_1 \ell \cdot \frac{o(o+1)}{2} + \lambda) \text{ bits.}$$

- The private key sk is a concatenation of byte strings representing $\text{seed}_{\text{sk}} \in \{0, 1\}^\lambda$ and seed_{pk} . Thus the size of secret key is 2λ bits.
- The signature σ is a concatenation of byte strings representing $\text{salt} \in \{0, 1\}^\lambda$ and $\mathbf{s}$. Thus the size of the signature is

$$(\lceil \log_2 q \rceil \cdot n \ell k + \lambda) \text{ bits.}$$

4.3 Basic functionalities

PRG expansion (PRG). For a PRG context initialized with seed and index t :

$$\text{PRG}(\text{seed}||t) := \text{RejSamp}(L, \tau_L, \text{pseudoXOF}(\tau, \text{seed}||\text{be16}(t))).$$

Here L denotes the number of output elements in $\mathbb{F}_q$, τ_L is the number of rejection sampling trials (each drawing a $[0, q-1]$ value from $[0, 2^{\lceil \log q \rceil} - 1]$) with failure probability bounded by $2^{-\lambda}$, and $\text{be16}(t)$ denotes the 2-byte big-endian encoding of the integer t .

Message digest (Expand_μ).

$$\text{Expand}_\mu(\text{seed}_{\text{pk}} \| M) := \text{pseudohash}(512, \text{seed}_{\text{pk}} \| M) \in \{0, 1\}^{512}.$$

Challenge hash (Hash).

$$\text{Hash}(\mu \| \text{salt}) := \text{RejSamp}(m_2, \tau_{m_2}, \text{pseudoXOF}(\tau_{m_2}, \mu \| \text{salt})) \in \mathbb{F}_q^{m_2}.$$

Linear map (π_ℓ, π_1). For $\bar{x} \in \mathbb{F}_{q^\ell}$,

$$\begin{aligned} \pi_\ell : \mathbb{F}_{q^\ell} &\rightarrow \mathbb{F}_q^\ell, & \bar{x} &\mapsto \Phi_{\bar{x}}[0, :] \\ \pi_1 : \mathbb{F}_{q^\ell} &\rightarrow \mathbb{F}_q, & \bar{x} &\mapsto \Phi_{\bar{x}}[0, 0] \end{aligned}$$

For vector $\bar{\mathbf{x}}$ over $\mathbb{F}_{q^\ell}$, $\pi_\ell(\bar{\mathbf{x}})[i] := \pi_\ell(\bar{\mathbf{x}}[i])$ and $\pi_1(\bar{\mathbf{x}})[i] := \pi_1(\bar{\mathbf{x}}[i])$.

$\mathbb{F}_{q^\ell}$ **packing** ($\text{Pack}_{\lceil \log q \rceil}$). The representation of $\mathbb{F}_{q^\ell}$ consists of ℓ $\mathbb{F}_q$ -coefficients, where each coefficient is encoded in $\lceil \log q \rceil$ -bit.

Algorithm 1 TSUOV KeyGen

Input: para

Output: (cpk, csk)

```

1: procedure COMPACTKEYGEN(para =  $(\lambda, n, o, m_1, m_2, \ell, k, q)$ )
2:    $v = n - o$ 
3:    $\text{seed}_{\text{sk}} \leftarrow \{0, 1\}^\lambda$ 
4:    $\bar{\mathbf{S}}' := \text{PRG}(\text{seed}_{\text{sk}}) \in \mathbb{F}_{q^\ell}^{v \times o}$ 
5:    $\text{seed}_{\text{pk}} \leftarrow \{0, 1\}^\lambda$ 
6:   for  $i = 0$  to  $m_1 - 1$  do
7:      $\bar{\mathbf{P}}_{i,1} = \text{PRG}(\text{seed}_{\text{pk}} \| 2i)$   $\triangleright$  symmetric  $\bar{\mathbf{P}}_{i,1} \in \mathbb{F}_{q^\ell}^{v \times v}$ 
8:      $\bar{\mathbf{P}}_{i,2} = \text{PRG}(\text{seed}_{\text{pk}} \| 2i + 1)$   $\triangleright \bar{\mathbf{P}}_{i,2} \in \mathbb{F}_{q^\ell}^{v \times o}$ 
9:      $\bar{\mathbf{P}}_{i,3} = (-\bar{\mathbf{S}}'^\top \bar{\mathbf{P}}_{i,1} + \bar{\mathbf{P}}_{i,2}^\top) \bar{\mathbf{S}}' + \bar{\mathbf{S}}'^\top \bar{\mathbf{P}}_{i,2}$   $\triangleright \bar{\mathbf{P}}_{i,3} \in \mathbb{F}_{q^\ell}^{o \times o}$  is symmetric
10:  end for
11:   $\text{cpk} = \text{seed}_{\text{pk}} \| \text{Pack}_{\lceil \log q \rceil}(\{\bar{\mathbf{P}}_{i,3}\}_{i \in [m_1]})$   $\triangleright$  upper triangle only
12:   $\text{csk} = \text{seed}_{\text{sk}} \| \text{seed}_{\text{pk}}$ 
13:  return (cpk, csk)
14: end procedure

```

Algorithm 2 TSUOV Sign

Input: (para, csk, M)

Output: sig = salt || Pack $_{\lceil \log q \rceil}(\bar{s})$

```

1:  $\mu = \text{Expand}_{\mu}(\text{seed}_{\text{pk}} \| M)$ 
2:  $\bar{\mathbf{S}}' = \text{PRG}(\text{seed}_{\text{sk}}) \in \mathbb{F}_{q^\ell}^{v \times o}$ 
3:  $\text{seed}_v \leftarrow \{0, 1\}^\lambda$ 
4:  $\bar{\mathbf{v}}_0 \| \dots \| \bar{\mathbf{v}}_{k-1} = \text{PRG}(\text{seed}_v)$   $\triangleright \bar{\mathbf{v}}_i \in \mathbb{F}_{q^\ell}^v$ 
5: for  $r = 0$  to  $m_1 - 1$  do
6:    $\bar{\mathbf{P}}_{i,1} = \text{PRG}(\text{seed}_{\text{pk}} \| 2i)$   $\triangleright$  symmetric  $\bar{\mathbf{P}}_{i,1} \in \mathbb{F}_{q^\ell}^{v \times v}$ 
7:    $\bar{\mathbf{P}}_{i,2} = \text{PRG}(\text{seed}_{\text{pk}} \| 2i + 1)$   $\triangleright \bar{\mathbf{P}}_{i,2} \in \mathbb{F}_{q^\ell}^{v \times o}$ 
8:   for  $i = 0$  to  $k - 1$  do
9:      $\mathbf{M}_i[r, :] = \pi_\ell(-\bar{\mathbf{v}}_i^\top \bar{\mathbf{P}}_{i,1} \cdot \bar{\mathbf{S}}' + \bar{\mathbf{v}}_i^\top \bar{\mathbf{P}}_{i,2})$   $\triangleright \mathbf{M}_i \in \mathbb{F}_q^{m_1 \times lo}$ 
10:    for  $j = i$  to  $k - 1$  do
11:       $\mathbf{U}_{ij}[r] = \pi_1(\bar{\mathbf{v}}_i^\top \bar{\mathbf{P}}_{r,1} \bar{\mathbf{v}}_j)$   $\triangleright \mathbf{U}_{ij} \in \mathbb{F}_q^{m_1}$ 
12:    end for
13:  end for
14: end for
15:  $\text{ctr} = 0$ ,  $\mathbf{u} = \mathbf{0}_{m_2 \times 1}$ ,  $\mathbf{A}_i = \mathbf{0}_{m_2 \times lo} (i \in [k])$ 
16: for  $i = 0$  to  $k - 1$  do
17:   for  $j = k - 1$  to  $i$  do
18:      $\mathbf{u} = \mathbf{u} + \mathbf{E}_{\text{ctr}} \mathbf{U}_{ij}$ 
19:      $\mathbf{A}_i = \mathbf{A}_i + \mathbf{E}_{\text{ctr}} \mathbf{M}_j$ 
20:      $\mathbf{A}_j = \mathbf{A}_j + \mathbf{E}_{\text{ctr}} \mathbf{M}_i$ 
21:      $\text{ctr} = \text{ctr} + 1$ 
22:   end for
23: end for
24:  $\mathbf{L} := [\mathbf{A}_0, \dots, \mathbf{A}_{k-1}] \in \mathbb{F}_q^{m_2 \times k lo}$ 
25: while True do
26:    $\text{salt} \leftarrow \{0, 1\}^\lambda$ 
27:    $\mathbf{t} = \text{Hash}(\mu \| \text{salt}) \in \mathbb{F}_q^{m_2}$ 
28:   if  $\mathbf{Lx} = \mathbf{t} - \mathbf{u}$  is solvable then
29:     Solve  $\mathbf{Lx} = \mathbf{t} - \mathbf{u}$ , obtain  $\mathbf{x} \in \mathbb{F}_q^{k lo}$ 
30:     for  $i = 0$  to  $k - 1$  do
31:        $\bar{\mathbf{o}}_i = \pi_\ell^{-1}(\mathbf{x}[i lo : (i + 1) lo])$   $\triangleright \bar{\mathbf{o}}_i \in \mathbb{F}_{q^\ell}^o$ 
32:        $\bar{\mathbf{s}}_i = (\bar{\mathbf{v}}_i - \bar{\mathbf{S}}' \bar{\mathbf{o}}_i, \bar{\mathbf{o}}_i)$   $\triangleright \bar{\mathbf{s}}_i \in \mathbb{F}_{q^\ell}^n$ 
33:     end for
34:      $\bar{\mathbf{s}} = (\bar{\mathbf{s}}_0, \dots, \bar{\mathbf{s}}_{k-1})$   $\triangleright \bar{\mathbf{s}} \in \mathbb{F}_{q^\ell}^{nk}$ 
35:     return sig = salt || Pack $_{\lceil \log q \rceil}(\bar{\mathbf{s}})$ 
36:   end if
37: end while

```

Algorithm 3 TSUOV Verify

Input: (para, cpk, M , sig)**Output:** $\{0, 1\}$

```
1: Unpack sig as salt  $\parallel \bar{s}$ 
2: Unpack cpk as seedpk  $\parallel \{\mathbf{P}_{i,3}\}_{i \in [m_1]}$ 
3:  $\mu \leftarrow \text{Expand}_\mu(\text{seed}_{\text{pk}} \parallel M)$ 
4:  $\mathbf{t} \leftarrow \text{Hash}(\mu \parallel \text{salt}) \in \mathbb{F}_q^{m_2}$ 
5: for  $r = 1$  to  $m_1$  do
6:    $\bar{\mathbf{P}}_{r,1} = \text{PRG}(\text{seed}_{\text{pk}} \parallel 2r)$   $\triangleright$  symmetric  $\bar{\mathbf{P}}_{r,1} \in \mathbb{F}_q^{v \times v}$ 
7:    $\bar{\mathbf{P}}_{r,2} = \text{PRG}(\text{seed}_{\text{pk}} \parallel 2r + 1)$   $\triangleright \bar{\mathbf{P}}_{r,2} \in \mathbb{F}_q^{v \times o}$ 
8:   for  $i = 0$  to  $k - 1$  do
9:      $\mathbf{c} = \pi_\ell \left( \bar{\mathbf{s}}_i^\top \begin{bmatrix} \bar{\mathbf{P}}_{r,2} & \bar{\mathbf{P}}_{r,2} \\ \bar{\mathbf{P}}_{r,2}^\top & \bar{\mathbf{P}}_{r,3} \end{bmatrix} \right)$   $\triangleright \mathbf{c} \in \mathbb{F}_q^n$ 
10:    for  $j = i$  to  $k - 1$  do
11:       $\mathbf{U}_{ij}[r] = \mathbf{c} \cdot \pi_\ell(\mathbf{s}_j)$   $\triangleright \mathbf{U}_{ij} \in \mathbb{F}_q^{m_1}$ 
12:    end for
13:  end for
14: end for
15:  $\text{ctr} = 0, \mathbf{y} = \mathbf{0} \in \mathbb{F}_q^{m_2}$ 
16: for  $i = 0$  to  $k - 1$  do
17:   for  $j = k - 1$  to  $i$  do
18:      $\mathbf{y} \leftarrow \mathbf{y} + \mathbf{E}_{\text{ctr}} \pi_1(\mathbf{U}_{ij})$ 
19:      $\text{ctr} := \text{ctr} + 1$ 
20:   end for
21: end for
22: return ( $\mathbf{y} == \mathbf{t}$ )
```

4.4 Parameter sets

For the parameters $q = 31$ and $\ell = 2$, we choose the symmetric matrix $\Phi = \begin{pmatrix} 1 & 8 \\ 8 & -1 \end{pmatrix}$ with its irreducible characteristic polynomial $f(x) = x^2 - 3$. Let $g(z) \in \mathbb{F}_q[z]$ be an irreducible polynomial of degree m_2 . We define $\mathbf{E}_i \in \mathbb{F}_q^{m_2 \times m_1}$ as the submatrix comprising the first m_1 columns of the matrix representation of z^i in the quotient ring $\mathbb{F}_q[z]/\langle g(z) \rangle$. The specific choices of $g(z)$ for different security levels are provided in Table 4.4.

Scheme ICCS level(λ)	TSUOV_128 128	TSUOV_256 256	TSUOV_512 512
(q, ℓ)	$(31, 2)$	$(31, 2)$	$(31, 2)$
n	64	119	230
o	4	5	12
m_1	61	114	218
k	11	16	10
m_2	88	160	240
$g(z)$	$z^{88} - z^2 - 3z - 4$	$z^{160} - z^2 - 4z - 6$	$z^{240} - z^2 - 12z - 15$
secret key (bytes)	32	64	128
public key (bytes)	778	2170	21319
signature (bytes)	896	2412	2939

Table 1: Parameter sets for TSUOV.

5 Performance Analysis

5.1 Reference implementation

The performance evaluation of the TSUOV reference implementation is conducted on a compliant environment as specified in the self-assessment guideline. The detailed software and hardware configurations are specified below to ensure the reproducibility of the benchmark results.

Software Configuration. The scheme is developed in ISO C99 and compiled using gcc (version 9.4.0) with the flags `-std=c99 -Wpedantic -Wall -Wextra -O2`. The operating system is Ubuntu 20.04.6 LTS running on Windows Subsystem for Linux (WSL2, kernel 6.18.33.2-microsoft-standard-WSL2). The build system uses CMake 3.16.3.

Cryptographic Primitives. For hash functions and pseudorandom number generation (PRNG), the underlying hash SM3 is utilized.

Hardware Platform. The benchmarks are executed on a 64-bit x86-64 platform (12th Gen Intel Core i9-12900H) with 15 GB RAM. Although the processor supports advanced vector instructions such as AVX2, this reference C implementation does not explicitly utilize SIMD-level optimizations.

Benchmarking Methodology. Each operation is run 500 times (≥ 100); signing uses a fixed 64-byte message. The reported CPU cycle counts represent the average values derived from 500 independent trials for each parameter set, ensuring a statistically stable performance profile.

Table 2 reports the average cycle counts and corresponding throughput (opt/s).

SL	KeyGen	Sign	Verify
	Cycles (opt/s)	Cycles (opt/s)	Cycles (opt/s)
TSUOV_128	8,870,006 (329.02)	17,038,245 (171.28)	14,871,690 (196.24)
TSUOV_256	62,930,059 (46.38)	138,743,481 (21.03)	125,232,329 (23.30)
TSUOV_512	983,120,817 (2.97)	841,661,427 (3.47)	889,272,156 (3.28)

Table 2: Average CPU cycles and throughput (operations per second) of reference implementation over 500 runs.

5.2 Performance-optimized implementation

The performance-optimized implementation leverages AVX2 vector instructions to accelerate the TSUOV operations. The evaluation is conducted on the same hardware platform as the reference implementation, with the compiler configuration tuned for SIMD-level optimization.

Software Configuration. The scheme is developed in ISO C99 and compiled using gcc (version 9.4.0) with the flags `-O3 -march=x86-64 -mavx2 -mtune=native -flto -fomit-frame-pointer -std=c99 -Wpedantic -Wall -Wextra`. The operating system is Ubuntu 20.04.6 LTS running on Windows Subsystem for Linux (WSL2, kernel 6.18.33.2-microsoft-standard-WSL2). The build system uses CMake 3.16.3.

Cryptographic Primitives. For hash functions and pseudorandom number generation (PRNG), the underlying hash SM3 is utilized.

Hardware Platform. The benchmarks are executed on a 64-bit x86-64 platform (12th Gen Intel Core i9-12900H) with 15 GB RAM. The processor supports advanced vector instructions including AVX2, which are explicitly utilized in this optimized implementation.

Benchmarking Methodology. Each operation is run 500 times (≥ 100); signing uses a fixed 64-byte message. The reported CPU cycle counts represent the average values derived from 500 independent trials for each parameter set, ensuring a statistically stable performance profile.

Table 3 reports the average cycle counts and corresponding throughput (opt/s). A more optimized constant-time implementation is left for future work.

Scheme	KeyGen	Sign	Verify
	Cycles (opt/s)	Cycles (opt/s)	Cycles (opt/s)
TSUOV_128	5,206,568 (560.52)	6,243,822 (467.40)	5,505,839 (530.05)
TSUOV_256	32,983,645 (88.48)	38,956,734 (74.91)	36,580,096 (79.78)
TSUOV_512	438,368,587 (6.66)	361,843,558 (8.07)	412,087,359 (7.08)

Table 3: Average CPU cycles and throughput (operations per second) of AVX2 implementation over 500 runs.

The profiling results reveal that pseudorandom number generation (PRG expansion) constitutes the absolute performance bottleneck in the optimized implementation, accounting for over 80% of the total execution cycles across all security levels and opera-

tions. Specifically, the PRG expansion dominates KeyGen at approximately 95%, Sign at 82–95% (with the proportion increasing from the 128-bit to the 512-bit security level), and Verify at 87–96%. The remaining computational workload—primarily matrix and vector-matrix multiplications—contributes only a marginal fraction of roughly 4–5% in KeyGen and 4–13% in Sign and verify.

A critical observation is that the PRG expansion in Sign and Verify serves to reconstruct the private and public key material from a compact seed. Since the key pair is generated only once and subsequently reused for multiple sign/verify operations, this expansion step can be precomputed and cached after the initial KeyGen, or alternatively upon the first Sign or first Verify operation. In practical deployment scenarios, the dominant PRG cost is therefore effectively amortized over numerous sign/verify invocations, and the per-operation latency of Sign and Verify is dominated by the lightweight arithmetic components rather than the PRG expansion.

6 Security Analysis

6.1 Hard Problems underlying TSUOV

The first hard problem is the core of hardness for UOV-like schemes. We have

Definition 6.1 (OV problem). For $\mathbf{O} \in \mathbb{F}_q^{(n-o) \times o}$, the OV problem asks to distinguish a random multivariate quadratic map that vanishes on the rowspace of $[-\mathbf{O}^\top \quad \mathbf{I}_o]$ from a random multivariate quadratic map.

Denote $\text{MQ}_{n,m,q}$ to be the set of quadratic maps from $\mathbb{F}_q^n$ to $\mathbb{F}_q^m$, and denote $\text{MQ}_{n,m,q}(\mathbf{O})$ to be subset of $\text{MQ}_{n,m,q}$ that vanishes on $[-\mathbf{O}^\top \quad \mathbf{I}_o]$.

We say that for $\mathcal{A}$ an OV distinguisher algorithm, its advantage is

$$\text{Adv}_{n,m,o,q}^{\text{OV}}(\mathcal{A}) = \left| \Pr[\mathcal{A}(\mathcal{P}) = 1 \mid \mathcal{P} \xleftarrow{\$} \text{MQ}_{n,m,q}] - \Pr\left[\mathcal{A}(\mathcal{P}) = 1 \mid \begin{matrix} \mathbf{O} \xleftarrow{\$} \mathbb{F}_q^{(n-o) \times o} \\ \mathcal{P} \xleftarrow{\$} \text{MQ}_{n,m,q}(\mathbf{O}) \end{matrix} \right] \right|$$

Besides traditional OV problem, we have a new problem targeted at TSUOV, that says tensoring the map makes it resistant to multi-target preimage attack.

Definition 6.2 (Multi-Target Tensor MQ problem). For some matrices $\{\mathbf{E}^{(s,t)}\}_{s \in [m_2], t \in [m_1]}$ and symmetric irreducible matrix $\mathbf{S}$, given random quadratic map $\mathcal{P}$ and access to an unbounded number of random targets $\mathbf{t}_i \in \mathbb{F}_q^m$ for $i \in \mathbb{N}$, the multi-target Tensor MQ problem asks to compute $(I, \mathbf{s})$ such that $\tilde{\mathcal{P}}(\mathbf{s}) = \mathbf{t}_I$, where the construction of $\tilde{\mathcal{P}}$ follows Section 3

For an adversary $\mathcal{A}$ the advantage of $\mathcal{A}$ against this problem is

$$\text{Adv}_{\{\mathbf{E}^{(s,t)}\}, \mathbf{S}, n, m, k, q}^{\text{MTTSMQ}}(\mathcal{A}) = \Pr\left[\tilde{\mathcal{P}}(\mathbf{s}) = \mathbf{t}_I \mid \mathcal{P} \text{ is quadratic, } \{\mathbf{t}_i\} \in \mathbb{F}_q^{m \times N}, (I, \mathbf{s}) \leftarrow \mathcal{A}^{\mathbf{t}_i}(\mathcal{P})\right]$$

6.2 Security Proof

Lemma 6.3 (Conditional law of RejSamp). *Let $\rho \xleftarrow{\$} \{0, 1\}^\tau$ uniformly, and for $\mathbf{t} \in \mathbb{F}_q^{m_2}$ define*

$$R_{\mathbf{t}} := \{\rho \in \{0, 1\}^\tau \mid \text{RejSamp}(\rho) = \mathbf{t}\}.$$

For every $\mathbf{t}$ with $R_{\mathbf{t}} \neq \emptyset$ and every $\rho_0 \in R_{\mathbf{t}}$, we have

$$\Pr[\rho = \rho_0 \mid \text{RejSamp}(\rho) = \mathbf{t}] = \frac{1}{|R_{\mathbf{t}}|}.$$

Proof. By Bayes' rule and uniformity of ρ over $\{0, 1\}^\tau$,

$$\Pr[\rho = \rho_0 \mid \text{RejSamp}(\rho) = \mathbf{t}] = \frac{\Pr[\rho = \rho_0 \wedge \text{RejSamp}(\rho) = \mathbf{t}]}{\Pr[\text{RejSamp}(\rho) = \mathbf{t}]} = \frac{2^{-\tau}}{|R_{\mathbf{t}}| \cdot 2^{-\tau}} = \frac{1}{|R_{\mathbf{t}}|}.$$

□

Lemma 6.4 (Fiber laws of the uniform-preimage simulator). *For $\mathbf{t} \in \mathbb{F}_q^{m_2}$ define the fiber*

$$\mathcal{F}_{\mathbf{t}} := \{\mathbf{s} \in \mathbb{F}_q^{k\ell n} \mid \tilde{\mathcal{P}}(\mathbf{s}) = \mathbf{t}\}.$$

Consider the simulator distribution

$$\mathbf{s} \xleftarrow{\$} \mathbb{F}_q^{k\ell n}, \quad \mathbf{t} \leftarrow \tilde{\mathcal{P}}(\mathbf{s}).$$

Then for every $\mathbf{t}_0$ with $\mathcal{F}_{\mathbf{t}_0} \neq \emptyset$ and every $\mathbf{s}_0 \in \mathcal{F}_{\mathbf{t}_0}$:

1. $\Pr[\mathbf{s} = \mathbf{s}_0 \mid \mathbf{t} = \mathbf{t}_0] = 1/|\mathcal{F}_{\mathbf{t}_0}|,$
2. $\Pr[\mathbf{t} = \mathbf{t}_0] = |\mathcal{F}_{\mathbf{t}_0}|/q^{k\ell n}.$

Proof. For (1), conditioning on $\mathbf{t} = \mathbf{t}_0$ is equivalent to conditioning on $\mathbf{s} \in \mathcal{F}_{\mathbf{t}_0}$, and $\mathbf{s}$ is uniform over $\mathbb{F}_q^{k\ell n}$, so it is uniform over this subset.

For (2),

$$\Pr[\mathbf{t} = \mathbf{t}_0] = \Pr[\tilde{\mathcal{P}}(\mathbf{s}) = \mathbf{t}_0] = \frac{|\mathcal{F}_{\mathbf{t}_0}|}{q^{k\ell n}}.$$

□

Lemma 6.5 (Characterization by fiber laws). *Let random variables $(\mathbf{S}, \mathbf{T})$ satisfy, for every $\mathbf{t}_0$ with $\mathcal{F}_{\mathbf{t}_0} \neq \emptyset$,*

$$\Pr[\mathbf{S} = \mathbf{s}_0 \mid \mathbf{T} = \mathbf{t}_0] = \frac{1}{|\mathcal{F}_{\mathbf{t}_0}|} \quad \text{for all } \mathbf{s}_0 \in \mathcal{F}_{\mathbf{t}_0},$$

and

$$\Pr[\mathbf{T} = \mathbf{t}_0] = \frac{|\mathcal{F}_{\mathbf{t}_0}|}{q^{k\ell n}}.$$

Then $(\mathbf{S}, \mathbf{T})$ has the same law as the simulator in Lemma 6.4.

Proof. Let $(\mathbf{s}_0, \mathbf{t}_0)$ be arbitrary. If $\mathbf{s}_0 \notin \mathcal{F}_{\mathbf{t}_0}$ then $\Pr[\mathbf{S} = \mathbf{s}_0, \mathbf{T} = \mathbf{t}_0] = 0$. If $\mathbf{s}_0 \in \mathcal{F}_{\mathbf{t}_0}$, then

$$\Pr[\mathbf{S} = \mathbf{s}_0, \mathbf{T} = \mathbf{t}_0] = \Pr[\mathbf{T} = \mathbf{t}_0] \Pr[\mathbf{S} = \mathbf{s}_0 \mid \mathbf{T} = \mathbf{t}_0] = \frac{|\mathcal{F}_{\mathbf{t}_0}|}{q^{k\ell n}} \cdot \frac{1}{|\mathcal{F}_{\mathbf{t}_0}|} = q^{-k\ell n}.$$

These are exactly the point probabilities of the simulator law.

□

Lemma 6.6 (Uniformity of SAMPLESOLUTION). *Let $\mathbf{A} \in \mathbb{F}_q^{m \times ko}$ and $\mathbf{y} \in \mathbb{F}_q^m$. Write*

$$\mathcal{S}_{\mathbf{A}, \mathbf{y}} := \{\mathbf{x} \in \mathbb{F}_q^{ko} \mid \mathbf{A}\mathbf{x} = \mathbf{y}\}.$$

Assume $\text{rank}(\mathbf{A}) = m$ and $\mathcal{S}_{\mathbf{A}, \mathbf{y}} \neq \emptyset$. If $\mathbf{r} \xleftarrow{\$} \mathbb{F}_q^{ko}$ uniformly and $\mathbf{x} \leftarrow \text{SAMPLESOLUTION}(\mathbf{A}, \mathbf{y}, \mathbf{r})$, then conditioned on success, $\mathbf{x}$ is uniform over $\mathcal{S}_{\mathbf{A}, \mathbf{y}}$.

Proof. The row-echelon step determines a fixed pivot structure of $\mathbf{A}$ and expresses each pivot coordinate as an affine function of the free coordinates. Since the algorithm starts from a uniformly random $\mathbf{r}$ and uses back-substitution only to correct $\mathbf{r}$ into a solution, the induced map from the free coordinates to $\mathcal{S}_{\mathbf{A}, \mathbf{y}}$ is an affine bijection. Therefore each solution is hit by the same number of $\mathbf{r}$ values, and the output is uniform over $\mathcal{S}_{\mathbf{A}, \mathbf{y}}$. $\square$

Lemma 6.7 (Fiber-uniformity from uniform solution sampling). *Fix (pk, μ) and one successful invocation of SIGN. For each target $\mathbf{t}$, let*

$$\mathcal{X}_{\mathbf{t}} := \{\mathbf{x} \in \mathbb{F}_q^{k\ell o} \mid \mathbf{A}\mathbf{x} = \mathbf{t} - \mathbf{u}\}.$$

Assume the map from $\mathbf{x}$ to output signature vector $\mathbf{s}$ induced by the final loop of SIGN is a bijection from $\mathcal{X}_{\mathbf{t}}$ onto $\mathcal{F}_{\mathbf{t}}$. Then for every $\mathbf{t}_0$ with $\mathcal{F}_{\mathbf{t}_0} \neq \emptyset$ and every $\mathbf{s}_0 \in \mathcal{F}_{\mathbf{t}_0}$,

$$\Pr[\mathbf{s} = \mathbf{s}_0 \mid \mathbf{t} = \mathbf{t}_0, \text{success}] = \frac{1}{|\mathcal{F}_{\mathbf{t}_0}|}.$$

Proof. Under the stated bijection, each $\mathbf{s}_0 \in \mathcal{F}_{\mathbf{t}_0}$ corresponds to a unique preimage $\mathbf{x}_0 \in \mathcal{X}_{\mathbf{t}_0}$. Since $\mathbf{x}$ is uniform on $\mathcal{X}_{\mathbf{t}_0}$,

$$\begin{aligned} \Pr[\mathbf{s} = \mathbf{s}_0 \mid \mathbf{t} = \mathbf{t}_0, \text{success}] &= \Pr[\text{SAMPLESOLUTION}(\mathbf{A}, \mathbf{t}_0 - \mathbf{u}, \mathbf{r}) = \mathbf{x}_0 \mid \mathbf{t} = \mathbf{t}_0, \text{success}] \\ &= \frac{1}{|\mathcal{X}_{\mathbf{t}_0}|} \\ &= \frac{1}{|\mathcal{F}_{\mathbf{t}_0}|}. \end{aligned} \tag{35}$$

$\square$

Lemma 6.8 (Fiber-proportional target law from regular trials). *Fix (pk, μ) and consider the internal salt trials in SIGN before the first success. Assume:*

1. *each trial samples a target $\mathbf{t}$ uniformly from $\mathbb{F}_q^{m_2}$;*
2. *there exists a constant $c > 0$ such that the per-trial success probability conditioned on target $\mathbf{t}$ equals $c|\mathcal{F}_{\mathbf{t}}|$.*

Then for every $\mathbf{t}_0$,

$$\Pr[\mathbf{t} = \mathbf{t}_0 \mid \text{success}] = \frac{|\mathcal{F}_{\mathbf{t}_0}|}{q^{k\ell n}}.$$

Proof. Let $\pi(\mathbf{t}) := q^{-m_2}$ be the target law of one trial and let $p(\mathbf{t}) := c|\mathcal{F}_{\mathbf{t}}|$ be the conditional success probability. The target distribution at the first successful trial is proportional to $\pi(\mathbf{t})p(\mathbf{t})$, hence

$$\Pr[\mathbf{t} = \mathbf{t}_0 \mid \text{success}] = \frac{\pi(\mathbf{t}_0)p(\mathbf{t}_0)}{\sum_{\mathbf{t}} \pi(\mathbf{t})p(\mathbf{t})} = \frac{|\mathcal{F}_{\mathbf{t}_0}|}{\sum_{\mathbf{t}} |\mathcal{F}_{\mathbf{t}}|}.$$

The fibers partition $\mathbb{F}_q^{k\ell n}$, so $\sum_{\mathbf{t}} |\mathcal{F}_{\mathbf{t}}| = q^{k\ell n}$. Substituting gives the claim. $\square$

Lemma 6.9 (Successful signing output law). *Fix $(\mathbf{pk}, \mu)$ and consider one successful invocation of `SIGN`. Assume the hypotheses of Lemma 6.7 and Lemma 6.8. Then the induced distribution of $(\mathbf{s}, \mathbf{t})$ is identical to the simulator law*

$$\mathbf{s} \xleftarrow{\$} \mathbb{F}_q^{k\ell n}, \quad \mathbf{t} \leftarrow \tilde{\mathcal{P}}(\mathbf{s}).$$

Proof. Lemma 6.7 gives the conditional-uniform fiber law, and Lemma 6.8 gives the fiber-proportional target law. Applying Lemma 6.5 yields the distribution identity. $\square$

Lemma 6.10. *Suppose there exists an adversary $\mathcal{A}$ that runs in time T against the EUF-CMA security of the TSUOV signature scheme in the random oracle model with parameters in Section 4.4, makes at most Q_h random-oracle queries and at most Q_s signing queries. Let Q'_s be an upper bound on the total number of internal salt trials performed by the honest signing oracle while answering those Q_s signing queries, and let $\varepsilon_{\text{RejSamp}}$ be an upper bound on the probability that one call to `RejSamp` fails. Then there exists an adversary $\mathcal{B}$ against the EUF-KOA security of the TSUOV signature scheme, such that*

$$\text{Adv}_{n,m,o,k,l,q}^{\text{EUF-CMA}}(\mathcal{A}) \leq \text{Adv}_{n,m,o,k,l,q}^{\text{EUF-KOA}}(\mathcal{B}) + Q'_s \varepsilon_{\text{RejSamp}} + (Q_h + Q'_s) Q'_s 2^{-\lambda} + (Q_s + 1)^2 2^{-8 \text{digest_bytes}},$$

and $\mathcal{B}$ runs in time

$$T + \mathcal{O}((Q_h + Q'_s) \text{poly}(n, m, o, k, l, q))$$

Proof. We first give a description of the final EUF-KOA adversary $\mathcal{B} = \mathcal{B}^{\mathcal{A}}$:

Given a public key $\mathbf{pk}$ (which contains $\text{seed}_{\mathbf{pk}}$), the adversary $\mathcal{B}$ runs $\mathcal{A}$ on input $\mathbf{pk}$ and simulates all oracle access of $\mathcal{A}$. It maintains a lazy table L indexed by pairs (μ, salt) , where random-oracle calls correspond to calls to $\text{KDF-SM3}(\mu \parallel \text{salt}, \cdot)$.

Random-oracle simulation: on input (μ, salt) , if (μ, salt) is already in L , return the stored value ρ ; otherwise sample $\rho \xleftarrow{\$} \{0, 1\}^\tau$ uniformly, store it in L , and return ρ .

Signing-oracle simulation: on input msg , compute $\mu \leftarrow \text{KDF-SM3}(\text{seed}_{\mathbf{pk}} \parallel \text{msg}, \text{digest_bytes})$, sample $\text{salt} \xleftarrow{\$} \{0, 1\}^\lambda$, and abort if (μ, salt) already appears in L . Otherwise sample $\mathbf{s} \xleftarrow{\$} \mathbb{F}_q^{k\ell n}$ uniformly, compute $\mathbf{t} = \tilde{\mathcal{P}}(\mathbf{s})$, then sample

- ρ uniformly from $R_{\mathbf{t}} := \{\rho \in \{0, 1\}^\tau \mid \text{RejSamp}(\rho) = \mathbf{t}\}$,
store $(\mu, \text{salt}) \mapsto \rho$ in L , and return $\text{Encode}(\mathbf{s}) \parallel \text{salt}$.

When $\mathcal{A}$ outputs a forged pair (msg^*, σ^*) , $\mathcal{B}$ outputs the same pair.

We now justify $\mathcal{B}$ by a sequence of games.

1. Let Game_0 be the real EUF-CMA game with $\mathcal{A}$. Then

$$\Pr[\text{Game}_0() = 1] = \text{Adv}_{n,o,m,k,\ell,q}^{\text{EUF-CMA}}(\mathcal{A}).$$

2. Let Game_1 be the same as Game_0 , except that the game aborts immediately if any bytestring-to-Decode step inside the honest signing oracle fails its corresponding RejSamp test. Since there are at most Q'_s such internal trials in total and each fails with probability at most $\varepsilon_{\text{RejSamp}}$, we get

$$\Pr[\text{Game}_0() = 1] \leq \Pr[\text{Game}_1() = 1] + Q'_s \varepsilon_{\text{RejSamp}}.$$

3. Let Game_2 be the same as Game_1 , except that random-oracle values are generated lazily through table L , and the game aborts on bad event Coll : during an internal salt trial of the honest signer, the sampled pair (μ, salt) is already present in L (because of one of the Q_h adversarial random-oracle queries or one of the previous internal salt trials). For each of at most Q'_s internal trials, collision probability is at most $(Q_h + Q'_s)2^{-\lambda}$. Therefore

$$\Pr[\text{Game}_1() = 1] \leq \Pr[\text{Game}_2() = 1] + (Q_h + Q'_s)Q'_s 2^{-\lambda}.$$

4. Let Game_3 be the same as Game_2 , except for the following change inside each successful honest-signing response: for the final output pair $(\mathbf{s}, \mathbf{t})$, replace the signer-generated representative ρ by an independently sampled value $\rho' \xleftarrow{\$} R_{\mathbf{t}}$, and program the table entry at (μ, salt) to ρ' .

Under $\neg \text{Coll}$, this point has not been answered before, so reprogramming is invisible to previous adversarial view. For fixed $(\mathbf{s}, \mathbf{t})$, Lemma 6.3 implies that in Game_2 the conditional law of ρ is already uniform over $R_{\mathbf{t}}$, exactly as in Game_3 . Hence

$$\Pr[\text{Game}_2() = 1] = \Pr[\text{Game}_3() = 1].$$

5. Let Game_4 be the same as Game_3 , except that signing queries are answered by the public-key simulator in the quoted description above (no secret key is used).

By Lemma 6.9, conditioned on successful completion, the pair $(\mathbf{s}, \mathbf{t})$ produced by honest signing has the same law as the simulator's rule $\mathbf{s} \xleftarrow{\$} \mathbb{F}_q^{k\ell n}$ and $\mathbf{t} = \tilde{\mathcal{P}}(\mathbf{s})$. Since both games sample ρ uniformly from $R_{\mathbf{t}}$, the full tuple $(\rho, \mathbf{s}, \mathbf{t})$ is identically distributed, and therefore

$$\Pr[\text{Game}_3() = 1] = \Pr[\text{Game}_4() = 1].$$

6. Let Game_5 be the same as Game_4 , except that the game aborts if there exists two distinct messages $\text{msg} \neq \text{msg}'$ such that their digest collide. Since each signing oracle call and the final verification call uses the digest of the message, at most $(Q_s + 1)^2$ pairs of messages can collide, each with probability at most $2^{-8\text{digest_bytes}}$. Therefore

$$\Pr[\text{Game}_4() = 1] \leq \Pr[\text{Game}_5() = 1] + (Q_s + 1)^2 2^{-8\text{digest_bytes}}.$$

7. Let Game_6 be the final EUF-KOA game played by $\mathcal{B}^A$ with the above simulation. Under the complement of the digest-collision bad event of Game_5 , freshness in message implies freshness in digest input to signing, so every successful forgery in Game_5 is also successful in Game_6 . Therefore

$$\Pr[\text{Game}_5() = 1] \leq \Pr[\text{Game}_6() = 1] = \text{Adv}_{n,o,m,k,\ell,q}^{\text{EUF-KOA}}(\mathcal{B}).$$

A combination of inequalities gives us the result. $\square$

Lemma 6.11. *Let $\mathcal{A}$ be an EUF-KOA adversary that runs in time T against the TSUOV signature in the random oracle model with parameters in Section 4.4. Then there exists an adversary $\mathcal{B}$ against the $\text{OV}_{n,m,o,q,l}$ problem, and an adversary $\mathcal{B}'$ against the $\text{MTTSMQ}_{n,m,k,q,l}$ problem, that both run in time $T + O((1 + Q_h) \text{poly}(n, m, k, q, \ell))$, such that*

$$\text{Adv}_{n,o,m,k,\ell,q}^{\text{EUF-KOA}}(\mathcal{A}) \leq \text{Adv}_{n,m,o,q}^{\text{OV}}(\mathcal{B}^A) + \text{Adv}_{\{\{\mathbf{E}^{(s,t)}\}, \mathbf{S}, n, m, k, q\}}^{\text{MTTSMQ}}(\mathcal{B}'^A)$$

Proof. We still use a sequence of games to do the proof.

1. We define Game_0 to be the EUF-KOA game played by $\mathcal{A}$. By definition we have

$$\Pr[\text{Game}_0() = 1] = \text{Adv}_{n,m,o,k,\ell,q}^{\text{EUF-KOA}}(\mathcal{A}).$$

2. Game_1 is the same as Game_0 , except that during key generation, the challenger chooses a uniform random $\mathcal{P} \in \text{MQ}_{n,m,q}$ instead of a $\mathcal{P}$ that vanishes on some oil space $\mathcal{O}$. The adversary $\mathcal{B}$ uses $\mathcal{P}$ to reconstruct the encoding $\text{P1}, \text{P2}, \text{P3}$ for $\{\overline{\mathbf{P}}_i^{(1)}\}_{i \in [m]}$, $\{\overline{\mathbf{P}}_i^{(2)}\}_{i \in [m]}$, $\{\overline{\mathbf{P}}_i^{(3)}\}_{i \in [m]}$, respectively. $\mathcal{B}$ then generates seed_{pk} using the key generation algorithm, and run $\mathcal{A}$ using $\text{pk} = (\text{seed}_{\text{pk}} \parallel \text{P3})$, where a faithful simulation of random oracle KDF-SM3 is operated such that it returns $\text{P1} \parallel \text{P2}$ on seed_{pk} and returns random bytes otherwise. This ensures that if $\mathcal{P}$ is UOV instance then $\mathcal{B}^A$ is exactly Game_0 while if $\mathcal{P}$ is random instance then $\mathcal{B}^A$ is Game_1 . Therefore

$$\text{Adv}_{n,m,o,q}^{\text{OV}}(\mathcal{B}^A) = |\Pr[\text{Game}_0() = 1] - \Pr[\text{Game}_1() = 1]|$$

3. The next game Game_2 is the game played by another adversary $\mathcal{B}'^A$. When $\mathcal{B}'$ is given a multivariate quadratic map $\mathcal{P}$ and oracle access to arbitrary many random targets $\{\mathbf{t}_i\}_{i \in \mathbb{N}}$, it simulates Game_1 faithfully except that it responds to the i -th

random query using $\mathbf{t}_i$ (with some truncation or addition to fit the output length). When $\mathcal{A}$ outputs a message-signature pair $(\text{msg}, (\text{salt}, \mathbf{s}))$, $\mathcal{B}'$ first checks if this signature is valid. If it is valid, then it must be queried during the process, say I -th random oracle query. $\mathcal{B}'$ then outputs $(I, \mathbf{s})$. Otherwise $\mathcal{B}'$ aborts. The view of $\mathcal{A}$ is identical in Game_1 and Game_2 , since $\mathcal{B}'$ simulates Game_1 perfectly. Therefore

$$\text{Adv}_{\{\mathbf{E}^{(s,t)}\}, \mathbf{S}, n, m, k, q}^{\text{MTTSMQ}}(\mathcal{B}'^{\mathcal{A}}) = \Pr[\text{Game}_1() = 1]$$

The proof is completed by using triangle inequalities. $\square$

Combining the above two lemmas, we get:

Theorem 6.12. *Suppose there exists an EUF-CMA adversary $\mathcal{A}$ that runs in time T against the TSUOV signature scheme in the random oracle model with parameters in Section 4.4, makes at most Q_h random-oracle queries and at most Q_s signing queries. Let Q'_s be an upper bound on the total number of internal salt trials performed by the honest signing oracle while answering those Q_s signing queries, and let $\varepsilon_{\text{RejSamp}}$ be an upper bound on the probability that one call to RejSamp fails. Then there exists an adversary $\mathcal{B}^{\mathcal{A}}$ against the OV_{n, m, o, q^l} problem, and an adversary $\mathcal{B}'^{\mathcal{A}}$ against the $\text{MTTSMQ}_{n, m, k, q, l}$ problem, that both run in time $T + O((1 + Q_h) \text{poly}(n, m, k, q, \ell))$, such that*

$$\begin{aligned} \text{Adv}_{n, m, o, k, l, q}^{\text{EUF-CMA}}(\mathcal{A}) &\leq \text{Adv}_{n, m, o, q}^{\text{OV}}(\mathcal{B}^{\mathcal{A}}) + \text{Adv}_{\{\mathbf{E}^{(s,t)}\}, \mathbf{S}, n, m, k, q}^{\text{MTTSMQ}}(\mathcal{B}'^{\mathcal{A}}) \\ &\quad + Q'_s \varepsilon_{\text{RejSamp}} + (Q_h + Q'_s) Q'_s 2^{-\lambda} + (Q_s + 1) 2^{-8 \text{digest.bytes}}. \end{aligned}$$

6.3 Analysis of known attacks

6.3.1 Forgery attack

Definition 6.13 (Tensor MQ problem). Let $\mathcal{P}^* = (p_1^*, \dots, p_{m_2}^*) : \mathbb{F}_q^{n\ell k} \rightarrow \mathbb{F}_q^{m_2}$ be the public map of our new scheme. Given $\mathbf{t} \in \mathbb{F}_q^{m_2}$, the Tensor MQ problem is to find $\mathbf{x} \in \mathbb{F}_q^{n\ell k}$ such that

$$\mathcal{P}^*(\mathbf{x}) = \mathbf{t}. \quad (36)$$

There are two method to solve the above equation. One is to regard Eq. (36) as a random system of m_2 quadratic equations in $n\ell k$ variables and solve it directly using general purpose algorithms for MQ problem.

The other is to exploit the special structure of Eq. (36) induced by the tensor product and solve it accordingly. Indeed, Eq. (36) can be written as

$$\mathcal{P}^*(\mathbf{x}_1, \dots, \mathbf{x}_k) = \sum_{i=1}^k \mathbf{E}_{ii} \mathcal{P}(\mathbf{x}_i) + \sum_{1 \leq i < j \leq k} \mathbf{E}_{ij} \mathcal{P}'(\mathbf{x}_i, \mathbf{x}_j) = \mathbf{t}, \quad (37)$$

where $\mathbf{E}_{ij}$ is the $m_2 \times m_1$ matrix whose (s, a) -entry is the (i, j) -entry of $\mathbf{E}^{sa}$, $\mathcal{P} = (p_1, \dots, p_{m_1}) : \mathbb{F}_q^{n\ell} \rightarrow \mathbb{F}_q^{m_1}$ is the UOV map defined by P_i and

$$\mathcal{P}'(\mathbf{x}_i, \mathbf{x}_j) = (p'_1(\mathbf{x}_i, \mathbf{x}_j), \dots, p'_{m_1}(\mathbf{x}_i, \mathbf{x}_j)) : \mathbb{F}_q^{n\ell} \times \mathbb{F}_q^{n\ell} \rightarrow \mathbb{F}_q^{m_1}$$

is a bilinear map defined by $\mathcal{P}'(\mathbf{x}_i, \mathbf{x}_j) = \mathcal{P}(\mathbf{x}_i + \mathbf{x}_j) - \mathcal{P}(\mathbf{x}_i) - \mathcal{P}(\mathbf{x}_j)$. Therefore, the tensor MQ problem can be regarded as the whipping MQ problem in MAYO.

In direct attack, the attacker needs to solve the tensor MQ problem Eq. (36) or Eq. (37) for a given $\mathbf{t} := \mathbf{Hash}(\mathbf{msg}||\mathbf{salt}) \in \mathbb{F}_q^{m_2}$.

Take randomly $\alpha_1, \dots, \alpha_k \in \mathbb{F}_q$ and $\delta_1, \dots, \delta_k \in \mathbb{F}_q^{n_\ell}$, and set $\mathbf{x}_i = \alpha_i \mathbf{x} + \delta_i$ for $i = 1, \dots, k$. Then Eq. (37) reduces to

$$\begin{aligned} \mathcal{P}^*(\mathbf{x}_1, \dots, \mathbf{x}_k) &= \sum_{i=1}^k \alpha_i^2 \mathbf{E}_{ii} \mathcal{P}(\mathbf{x}) + \sum_{1 \leq i < j \leq k} \alpha_i \alpha_j \mathbf{E}_{ij} \mathcal{P}'(\mathbf{x}, \mathbf{x}) + \text{linear parts of } \mathbf{x} \\ &= \left(\sum_{i=1}^k \alpha_i^2 \mathbf{E}_{ii} + 2 \sum_{1 \leq i < j \leq k} \alpha_i \alpha_j \mathbf{E}_{ij} \right) \mathcal{P}(\mathbf{x}) + \text{linear parts of } \mathbf{x} \\ &= \mathbf{E}_\alpha \cdot \mathcal{P}(\mathbf{x}) + \text{linear parts of } \mathbf{x} = \mathbf{t}, \end{aligned} \quad (38)$$

where

$$\mathbf{E}_\alpha := \sum_{i=1}^k \alpha_i^2 \mathbf{E}_{ii} + 2 \sum_{1 \leq i < j \leq k} \alpha_i \alpha_j \mathbf{E}_{ij}.$$

If $\text{rank } \mathbf{E}_\alpha < m_1$, then Eq. (38) will become a system of less than m_1 equations in n_ℓ variables. To prevent this case, we can fix $\mathbf{E}_{ij}$ or $\mathbf{E}^{sa}$ such that every nonzero $\mathbb{F}_q$ -linear combination of $\mathbf{E}_{ij}$ has full column rank m_1 .

When k and q are small, we can randomly select $\mathbf{E}_{ij}$ (or $\mathbf{E}^{sa}$) and check whether every nonzero $\mathbb{F}_q$ -linear combination of $\mathbf{E}_{ij}$ has full column rank m_1 . However, when k and q are large, this method becomes inefficient. In the following, we provide a method to construct $\mathbf{E}_{ij}$ such that every nonzero $\mathbb{F}_q$ -linear combination of $\mathbf{E}_{ij}$ has full column rank m_1 .

Let $f \in \mathbb{F}_q[x]$ be an irreducible polynomial of degree m_2 , and $\mathbf{Z}$ be the companion matrix of f . Then we have

$$\mathbb{F}_q[\mathbf{Z}] := \{a_0 + a_1 \mathbf{Z} + \dots + a_{m_2-1} \mathbf{Z}^{m_2-1}\} \cong \mathbb{F}_{q^{m_2}}. \quad (39)$$

Set

$$\mathbf{E}_{ij} = \text{the left } m_1 \text{ columns of } \mathbf{Z}^{\frac{(i-1)(2k-i+2)}{2} + k - j} \in \text{Mat}_{m_2 \times m_1}(\mathbb{F}_q) \quad (40)$$

for any $1 \leq i \leq j \leq k$. Then every nonzero $\mathbb{F}_q$ -linear combination of the matrices $\mathbf{E}_{ij}$ has full column rank m_1 when

$$m_2 \geq \frac{k(k+1)}{2}. \quad (41)$$

Remark 6.14. Note that the above construction of $\mathbf{E}_{ij}$ comes from MAYO [4].

Claw finding attack An attacker can forge a signature for message by finding a salt and $\mathbf{s}$ such $\mathcal{P}^*(\mathbf{s}) = \mathbf{Hash}(\mathbf{msg}||\mathbf{salt})$ which is a claw-finding problem. Following [9], we estimate the number of gates required for this attack considering the cost of multiplication and addition in $\mathbb{F}_q$ as follows

$$2 \cdot \left(\frac{2}{q-1} \cdot q^{m_2} \cdot m_2 \cdot 2^{17} \cdot (2 \log_2(q)^2 + \log_2(q)) \right)^{\frac{1}{2}}. \quad (42)$$

On the other hand, if we fix $\alpha \neq 0$ in Eq. (38), we have $\text{rank } \mathbf{E}_\alpha = m_1$. Then Eq. (38) will become

$$\mathbf{E}_\alpha^{-1} \mathbf{E}_\alpha \mathcal{P}(\mathbf{x}) = \mathbf{E}_\alpha^{-1} \text{Hash}(\text{msg} || \text{salt}) \in \mathbb{F}_q^{m_1}, \quad (43)$$

where $\mathbf{E}_\alpha^{-1} \in \text{Mat}_{m_1 \times m_2}(\mathbb{F}_q)$ is a left inverse of $\mathbf{E}_\alpha$. $\mathbf{E}_\alpha^{-1}$ exists due to the fact that $\text{rank } \mathbf{E}_\alpha = m_1$. Therefore, the complexity of claw finding attack will become

$$2 \cdot \left(\frac{2}{q-1} \cdot q^{m_1} \cdot m_1 \cdot 2^{17} \cdot (2 \log_2(q)^2 + \log_2(q)) \right)^{\frac{1}{2}}. \quad (44)$$

Since in the design of parameters, we will set m_2 to be larger than m_1 , the complexity of claw finding attack will be dominated by Eq. (44).

Direct attack In direct attack, the attacker needs to solve the following system

$$\mathcal{P}^*(\mathbf{x}) = \mathbf{t}$$

for a given $\mathbf{t} \in \mathbb{F}_q^{m_2}$. As far as we know, there is currently no method that can leverage the structure of $\mathcal{P}^*$ to effectively solve the aforementioned equation. Therefore, we still treat it as a general stochastic quadratic equation with m_2 equations in $k\ell n$ variables. The complexity is

$$MQ(n\ell k, m_2, q). \quad (45)$$

On the other hand, if we assume that $\mathbf{x}_i = \alpha_i \mathbf{x} + \delta_i$ for some $\alpha_i \in \mathbb{F}_q$ and $\mathbf{x} \in \mathbb{F}_q^{n\ell}$, $\delta_i \in \mathbb{F}_q^{n\ell}$, then Eq. (37) reduces to Eq. (38). In this case,

$$\mathbf{E}_\alpha \cdot \mathcal{P}(\mathbf{x}) + \text{linear parts of } \mathbf{x} = \mathbf{t}. \quad (46)$$

Since $\text{rank } \mathbf{E}_\alpha = m_1$ when $\alpha = (\alpha_1, \dots, \alpha_k) \neq 0$, the above equation will become a system of m_1 quadratic equations and $m_2 - m_1$ linear equations in $n\ell$ variables. Therefore, the complexity will become

$$MQ(n\ell - (m_2 - m_1), m_1, q). \quad (47)$$

Furthermore, if we assume that $\mathbf{t}$ in Eq. (38) lies in the image of $\mathbf{E}_\alpha \cdot \mathbb{F}_q^{m_1}$. We need solve the system $\mathcal{P}(\mathbf{u}) = \mathbf{t}_1 \in \mathbb{F}_q^{m_1}$ with $n\ell$ variables and m_1 equations and the complexity will become

$$MQ(n\ell, m_1, q). \quad (48)$$

Therefore, we will use the minimal value of Eq. (45), Eq. (47) and Eq. (48) to estimate the complexity of direct attack.

6.3.2 Attacks on the Oil and Vinegar problem

One can try to attack by analyzing the UOV structure contained in the public map. If an adversary can recover the oil subspace of the UOV public map, then signing the message is as easy as doing this by the original signer.

In our scheme, though, there are many aspects to form UOV:

- $\text{UOV}(n, o, m_1, q^\ell)$, which uses $(\bar{\mathbf{F}}_i, \bar{\mathbf{S}}, \bar{\mathbf{P}}_i)$.
- $\text{UOV}(n\ell, o\ell, m_1, q)$, which uses $(\mathbf{F}_i, \mathbf{S}, \mathbf{P}_i)$.
- $\text{UOV}(n\ell k, o\ell k, m_2, q)$, which uses $(\mathbf{F}_i^*, \mathbf{I}_{k \times k} \otimes \mathbf{S}, \mathbf{P}_i^*)$

However we claim that it is sufficient to consider only the case $\text{UOV}(n, o, m_1, q^\ell)$ for complexity lower bound estimation. The reason is that there always exists a way to transform a $\text{UOV}(n, o, m_1, q^\ell)$ into other forms by following the construction in Section 3, but *not* vice versa.

Kipnis–Shamir attack In a $\text{UOV}(n, o, m, q)$ scheme, the Kipnis–Shamir attack [26] obtains the oil subspace $\mathcal{O}$ by choosing two invertible matrices $\mathbf{M}_1, \mathbf{M}_2$ from the set of linear combinations of public keys $\mathbf{P}_1, \dots, \mathbf{P}_m$ of the UOV scheme. Then it probabilistically recovers a part of the oil subspace $\mathcal{O}$ by finding the invariant subspace of $\mathbf{M}_1^{-1}\mathbf{M}_2$. The complexity of Kipnis–Shamir attack is estimated by

$$q^{n-2o} \cdot n^{2.8}$$

field multiplications. Grover’s algorithm [22] can be used to reduce the complexity for a quantum adversary to

$$q^{(n-2o)/2} \cdot n^{2.8}$$

field multiplications.

Then, the complexity of the Kipnis–Shamir attack for classical and quantum adversaries against $\text{UOV}(n, o, m_1, q^\ell)$ scheme is estimated by

$$q^{\ell(n-2o)} \cdot n^{2.8} \quad \text{and} \quad q^{\ell(n-2o)/2} \cdot n^{2.8}$$

field multiplications.

Reconciliation attack The reconciliation attack [14] for UOV is similar to the Kipnis–Shamir attack, trying to find an element of the oil subspace $\mathcal{O}$. The detailed attack for a $\text{UOV}(n, o, m, q)$ scheme can be decomposed into a series of steps:

1. Find an element $\mathbf{u} = (u_1, \dots, u_v, 0, \dots, 0, 1)^\top \in \mathbb{F}_q^n$ such that

$$\mathbf{u}^\top \cdot \mathbf{P}_i \cdot \mathbf{u} = 0 \in \mathbb{F}_q, \quad \text{for } i \in [m]. \quad (49)$$

There are m quadratic equations in v variables in Eq. (49).

2. Use the condition $\mathbf{u}^\top \cdot \mathbf{P}_i \cdot \mathbf{v} = 0$ to get m linear equations for other elements $\mathbf{v}$ of the oil subspace $\mathcal{O}$.

The complexity of reconciliation attack is mainly centered on solving Eq. (49). One can generalize the attack by using k linear independent oil elements $\mathbf{u}_j$ for $j \in [k]$ and solve for

$$\mathbf{u}_j^\top \cdot \mathbf{P}_i \cdot \mathbf{u}_{j'} = 0 \quad (50)$$

for $j, j' \in [k]$. We also point out that in the case of $v > m$, Eq. (49) will have a lot of spurious solutions not in the oil subspace $\mathcal{O}$. Therefore the complexity of the reconciliation attack is estimated by

$$\min_k q^{\max\{0, k(n-o) - \binom{k+1}{2}m\}} \text{MQ}(k(n-o), \binom{k+1}{2}m, q)$$

For our scheme, we find that reconciliation attack is most useful in the case $k = 1$ and our parameters always satisfy $n - o \leq m_1$, therefore the complexity is evaluated to be

$$\text{MQ}(n - o, m_1, q^\ell)$$

field multiplications.

Intersection attack In [6] the author proposed a new attack against $\text{UOV}(n, o, m, q)$ called the intersection attack. This attack inherits the idea of reconciliation attack, but tries to derive different oil subspace elements from one vector. The attack chooses two invertible matrices $\mathbf{M}_1, \mathbf{M}_2$ from the set of linear combinations of the public keys $\mathbf{P}_1, \dots, \mathbf{P}_m$. Suppose a vector $\mathbf{x} \in \mathbb{F}_q^n$ satisfies $\mathbf{M}_1^{-1}\mathbf{x}$ and $\mathbf{M}_2^{-1}\mathbf{x}$ lie in the oil subspace. Then $\mathbf{x}$ is a solution to the system

$$\begin{cases} (\mathbf{M}_1^{-1}\mathbf{x})^\top \cdot \mathbf{P}_i \cdot (\mathbf{M}_1^{-1}\mathbf{x}) = 0 \\ (\mathbf{M}_1^{-1}\mathbf{x})^\top \cdot \mathbf{P}_i \cdot (\mathbf{M}_2^{-1}\mathbf{x}) = 0 \\ (\mathbf{M}_2^{-1}\mathbf{x})^\top \cdot \mathbf{P}_i \cdot (\mathbf{M}_2^{-1}\mathbf{x}) = 0 \end{cases} \quad (51)$$

We consider two cases:

- In the case where $n < 3o$, such $\mathbf{x}$ exists since the image of oil subspace under $\mathbf{M}_1$ and $\mathbf{M}_2$ are both o -dimensional subspaces of the $(n - o)$ -dimensional orthogonal complement subspace of oil subspace. [6] pointed out that there should be 2 redundant equations in Eq. (51). One can impose $3o - n$ affine constraints on $\mathbf{x}$, so the attack is reduced to solving a quadratic system of $3m - 2$ equations in $n - (3o - n) = 2n - 3o$ variables. Therefore the complexity is estimated as

$$\text{MQ}(2n - 3o, 3m - 2, q)$$

field multiplications.

- In the case where $n \geq 3o$, such $\mathbf{x}$ may not exist nontrivially. The probability that such intersection exists is estimated to be approximately $q^{-n+3o-1}$. Moreover we cannot impose any affine constraints on $\mathbf{x}$, so the derived quadratic system contains $3m - 2$ equations and n variables. The complexity then is estimated as

$$q^{n-3o+1} \text{MQ}(n, 3m - 2, q)$$

field multiplications.

For our scheme, the parameters always satisfy $n \geq 3o$. Therefore the complexity is estimated to be

$$q^{\ell(n-3o+1)} \text{MQ}(n, 3m_1 - 2, q^\ell)$$

field multiplications.

Rectangular MinRank attack In [6] the author proposed the rectangular MinRank attack, which was later applied to QR-UOV by Furue and Ikematsu in [17]. For UOV(n, o, m, q) scheme, the rectangular MinRank attack tries to find a nonzero element $\mathbf{x} = (x_1, \dots, x_n)^\top$ of the oil space.

Suppose $\beta_i := \mathbf{P}_i \cdot \mathbf{x}$. Then there are o linear independent elements $\mathbf{y}_1, \dots, \mathbf{y}_o$ such that

$$\mathbf{y}_k^\top \cdot \beta_i = 0 \quad (52)$$

for any $i \in [m]$ and $k \in [o]$. This means that the matrix

$$\mathbf{B} := [\beta_1 \ \cdots \ \beta_m] \quad (53)$$

as a $n \times m$ matrix has a low rank of at most $n - o$. On the other hand notice that we can write

$$\mathbf{B} = x_1 \cdot \mathbf{P}'_1 + \cdots + x_n \cdot \mathbf{P}'_n \quad (54)$$

where $\mathbf{P}'_j$ is the *constant* matrix generated from concatenating j -th column of each public key matrix $\mathbf{P}_i$ horizontally in order. Combining with Eq. (49), we know that $\mathbf{x}$ is both a solution of the MinRank problem and that of the MQ problem, namely

$$\begin{cases} \mathbf{x}^\top \cdot \mathbf{P}_i \cdot \mathbf{x} = 0 & \text{for } i \in [m] \\ \text{rank}(x_1 \cdot \mathbf{P}'_1 + \cdots + x_n \cdot \mathbf{P}'_n) \leq n - o \end{cases} \quad (55)$$

In such case, $b_{\min}$ is the smallest value b such that the coefficient of t^b in

$$(1 - t^2)^m \cdot \sum_{i \geq 0} (M(i, 1) - R(i)) t^i$$

is non-positive.

Recently, based on [31], the authors in [35] extended MinRank attack. They proved that the target rank r can be chosen from a larger set. Indeed, they proved that

$$\dim(\Sigma_r \cap O) \geq D(r) := o - (m - r)(n - o - r)$$

where

$$\Sigma_r := \{\mathbf{x} \mid \mathbf{x}^\top \mathbf{P}_i \mathbf{x} = 0 \text{ for all } i, \text{ and } \text{rank}(x_1 \mathbf{P}'_1 + \cdots + x_n \mathbf{P}'_n) \leq r\}$$

Therefore the complexity of [35] should be

$$\min_{D(r) \geq 0} 3 \cdot \binom{m}{r}^2 \binom{n - D(r) + b_{\min}}{b_{\min}}^2 (r + 1)(n - D(r) + 1).$$

Where the last factor representing nonzero terms in each equation comes from the fact that since $\dim(\Sigma_r \cap O) > 0$, $D(r) - 1$ variables can be fixed to zero.

Wedge attack In [33], Ran introduced the *wedge attack*, a new key-recovery attack on UOV over fields of characteristic 2. Later, Jin et al. [25] generalized this approach to arbitrary characteristic by replacing the exterior algebra view point with a symmetric algebra formulation. For the notation adopted here, then framework of [25] corresponds to the case $\ell = 1$. Subsequently, Furue and Ikematsu [18] further extended this line of work to general p^ℓ -truncated polynomial rings

$$\mathbb{F}_q[x_1, \dots, x_n] / \langle x_1^{p^\ell}, \dots, x_n^{p^\ell} \rangle$$

thereby providing a unified treatment for all $1 \leq \ell \leq e$, where $q = p^e$.

For this truncated polynomial ring, the number of monomials of total degree d in n variables is

$$T(n, p^\ell, d) = \sum_{i=0}^{\min\{n, \lfloor d/p^\ell \rfloor\}} (-1)^i \binom{n}{i} \binom{d - ip^\ell + n - 1}{n - 1} \quad (56)$$

Using this truncated setting, one constructs a Macaulay submatrix $\mathbf{M}_d^{(p^\ell)}$. The key observation in [18] is that the oil space in $\text{UOV}(n, o, m, q)$ can be recovered from the kernel of this Macaulay matrix whenever

$$\dim \ker \mathbf{M}_d^{(p^\ell)} \leq T(o, p^\ell, d) \quad (57)$$

Moreover, the kernel dimension can be computed explicitly as

$$\dim \ker \mathbf{M}_d^{(p^\ell)} = \left\lfloor \frac{(1 + z + \dots + z^{p^\ell - 1})^n}{(1 + z^2 + \dots + z^{2(p^\ell - 1)})^m} \right\rfloor_d \quad (58)$$

where $[\cdot]_d$ denotes the coefficient of z^d in the corresponding power series.

Hence one defines

$$D := \min \left\{ 2 \leq d \leq o(p^\ell - 1) \mid \dim \ker \mathbf{M}_d^{(p^\ell)} \leq T(o, p^\ell, d) \right\} \quad (59)$$

For a $\text{UOV}(n, o, m, q)$ instance, the complexity of recovering the oil space by the wedge attack is then estimated by

$$\text{Comp}_{\text{Wedge}}(n, o, m, p^\ell) = 3 \binom{n+1}{2} \left(T(n, p^\ell, D) - T(o, p^\ell, D) + 1 \right)^2 \quad (60)$$

The reconciliation and intersection attacks described above admit natural wedge-based counterparts, which we call *wedge-reconciliation* and *wedge-intersection*, respectively. The reconciliation strategy attacks the given $\text{UOV}(n, o, m, q)$ instance directly, whereas the intersection strategy first transforms it into another UOV instance with modified parameters, which leads to a different complexity estimate. In both cases, the projection-down technique can be incorporated, and one may further optimize the attack by choosing suitable values of the projection parameter k and the truncation parameter ℓ .

Wedge-reconciliation In the reconciliation setting, one starts from the original UOV instance $\text{UOV}(n, o, m, q = p^e)$. After applying projection-down and choosing a truncation level ℓ , the resulting complexity is estimated by

$$\min_{\substack{1 \leq \ell \leq e \\ 0 \leq k < o}} \text{Comp}_{\text{Wedge}}(n, o - k, m, p^\ell) \quad (61)$$

Wedge-intersection In the intersection setting, the original $\text{UOV}(n, o, m, q = p^e)$ instance is first transformed into a new UOV instance:

- If $n < 3o$, the transformed instance has parameters

$$\text{UOV}(2n - 3o, 3o - n, 3m - 2, q)$$

and the corresponding complexity estimate is

$$\min_{\substack{1 \leq \ell \leq e \\ 0 \leq k < 3o - n}} \text{Comp}_{\text{Wedge}}(2n - 3o, 3o - n - k, 3m - 2, p^\ell) \quad (62)$$

- If $n \geq 3o$, the transformed instance becomes

$$\text{UOV}(n, 1, 3m - 2, q)$$

with probability $q^{-n+3o-1}$. In this case the corresponding complexity estimate becomes

$$q^{n-3o+1} \cdot \text{Comp}_{\text{Wedge}}(n - 1, 1, 3m - 2, p^\ell) \quad (63)$$

7 Advantages and Limitations

In this section we summarize the advantages and the limitations of our TSUOV in this submission.

7.1 Advantages

Significantly Reduced Public Key Size TSUOV achieves substantially smaller public key sizes compared to other UOV variants through its unified tensor-based framework. This represents a significant reduction in storage and transmission overhead, making the scheme particularly suitable for bandwidth-constrained applications.

Competitive Performance with Lattice-based Schemes When compared to standardized lattice-based schemes like Dilithium (ML-DSA), TSUOV offers superior key and signature size efficiency.

Computational Efficiency TSUOV maintains the computational efficiency inherent to UOV-based schemes, requiring only simple linear algebra operations over finite fields.

Unified Security Framework TSUOV integrates and generalizes the design principles of QR-UOV, MAYO, and SNOVA through a tensor product framework, providing security guarantees no weaker than any of these individual schemes. The scheme has been analyzed against all known attacks on multivariate quadratic systems, including claw-finding attacks, direct attacks (Hashimoto’s method with WXL), and UOV-specific attacks (Kipnis-Shamir, reconciliation, intersection, rectangular Minrank, and wedge attacks).

Flexible The unified design of TSUOV allows for flexible parameter selection and potential future enhancements. By adjusting the tensor product structure and the dimensions of the oil and vinegar spaces, it is possible to optimize for different security levels or application requirements without fundamentally altering the underlying scheme. This adaptability is a key advantage in the rapidly evolving landscape of post-quantum cryptography.

7.2 Limitations

Moderate Signature Size While TSUOV achieves substantial reductions in public key size compared to QR-UOV and MAYO, the signature size is moderately larger than some alternatives.

Complexity and Implementation Expertise While TSUOV maintains the fundamental simplicity of linear algebra operations, the unified tensor product framework requires more sophisticated understanding than individual UOV variants. Implementers need familiarity with both tensor product structures and extension field arithmetic. Careful attention to constant-time implementation is necessary to prevent timing side-channel attacks, particularly in the solving of linear systems and matrix operations.

Field Arithmetic Overhead TSUOV operations are conducted over extension fields $\mathbb{F}_{q^\ell}$, requiring more complex field arithmetic than base-field operations in standard UOV. This adds computational overhead in field element multiplication and inversion, although the AVX2 optimizations partially mitigate this cost. Implementations on platforms lacking SIMD support may experience reduced performance relative to simpler schemes.

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